

Composition driven prediction of thermoelectric figure of merit(ZT) using benchmarked Machine learning, Stacked ensemble Learning and similarity-guided inverse design

Sudarshan, Sai Jeevan, Shatrujit

Abstract:

Thermoelectric materials enable direct conversion of heat into electricity, and their efficiency is commonly evaluated using the dimensionless figure of merit (ZT). Experimental discovery of high performance thermoelectric materials remains expensive and time intensive due to the large compositional search space. In this work, a composition-driven material informatics framework was developed for ZT prediction and candidate generation using machine learning. Experimental thermoelectric data obtained from StarryData2.0 were transformed into physically informed descriptors using Pymatgen. Multiple regression and deep learning models were benchmarked, including CatBoost, Gradient Boosting Machine (GBM), ExtraTrees, LightGBM, XGBoost, and ZTNet. Two stacked ensemble architectures were additionally investigated.

Among all evaluated approaches, the ExtraTrees-LightGBM stacked ensemble achieved the strongest predictive performance with a test R^2 of 0.9202 under five-fold cross-validation. Residual analysis indicated low systematic bias and stable predictive behavior.

To extend prediction toward candidate exploration, a similarity-guided inverse design framework was implemented using seed selection, controlled perturbation, surrogate prediction, Tanimoto similarity filtering, and physically motivated validation constraints. Generated candidates maintained strong similarity to experimentally observed materials while satisfying predefined thermoelectric performance criteria.

Feature importance and SHAP analysis identified temperature as the dominant descriptor, followed by compositional and electronic descriptors including cohesive energy, oxidation-related descriptors, and electron-affinity measures.

This work demonstrates an integrated workflow combining benchmarking, interpretable prediction, and constrained inverse design for thermoelectric material exploration.

Introduction:

Prediction of zT values is one of the major problems of modern material science because traditional trial and error method, experimental methods are time consuming and very costly.

Thermoelectric materials convert thermal gradients directly into electrical energy. Their performance is commonly quantified using the dimensionless figure of merit (ZT), which integrates electrical conductivity, Seebeck coefficient, thermal conductivity, and operating temperature into a unified efficiency metric.

$$ZT = \frac{S^2 \sigma T}{\kappa}$$

S : Seebeck coefficient (V/K)

σ : Electrical conductivity (S/m or $1/\Omega \cdot m$)

T : Absolute temperature (K)

κ : Thermal conductivity ($W/m \cdot K$) 

S : Seebeck coefficient (V/K)

σ : Electrical conductivity (S/m or $1/\Omega \cdot m$)

T : Absolute temperature (K)

κ : Thermal conductivity ($W/m \cdot K$) 

This work proposes a composition-driven machine learning framework for thermoelectric material discovery through benchmarked regression modeling, ensemble learning and similarity-guided inverse design.

Methodology:

1. Dataset and Feature Engineering

Experimental data came from the Starrydata2.0 database. We used Pymatgen to do feature engineering. This helped create descriptors from chemical compositions for a physically meaningful model. Many redundant features, over 20 were removed while training model. These included parts of ZT like Seebeck and

conductivity....etc.This was done to prevent data leakage and make

the model more robust. The remaining features were then normalized.

Our dataset which is ready to be given to a ML model, have a 120 features and 32635 samples. All the features except composition are numerical features.

Then we visualized the data set using correlation heat map to PCA , to understand how features were related to each other.

Exploratory analysis suggested substantial nonlinear relationships among descriptors. Preliminary benchmarking using linear and nonlinear models further confirmed that tree-based approaches were better suited for this dataset.

Results of data modeling : [R-square values]

Linear regression:- 0.50

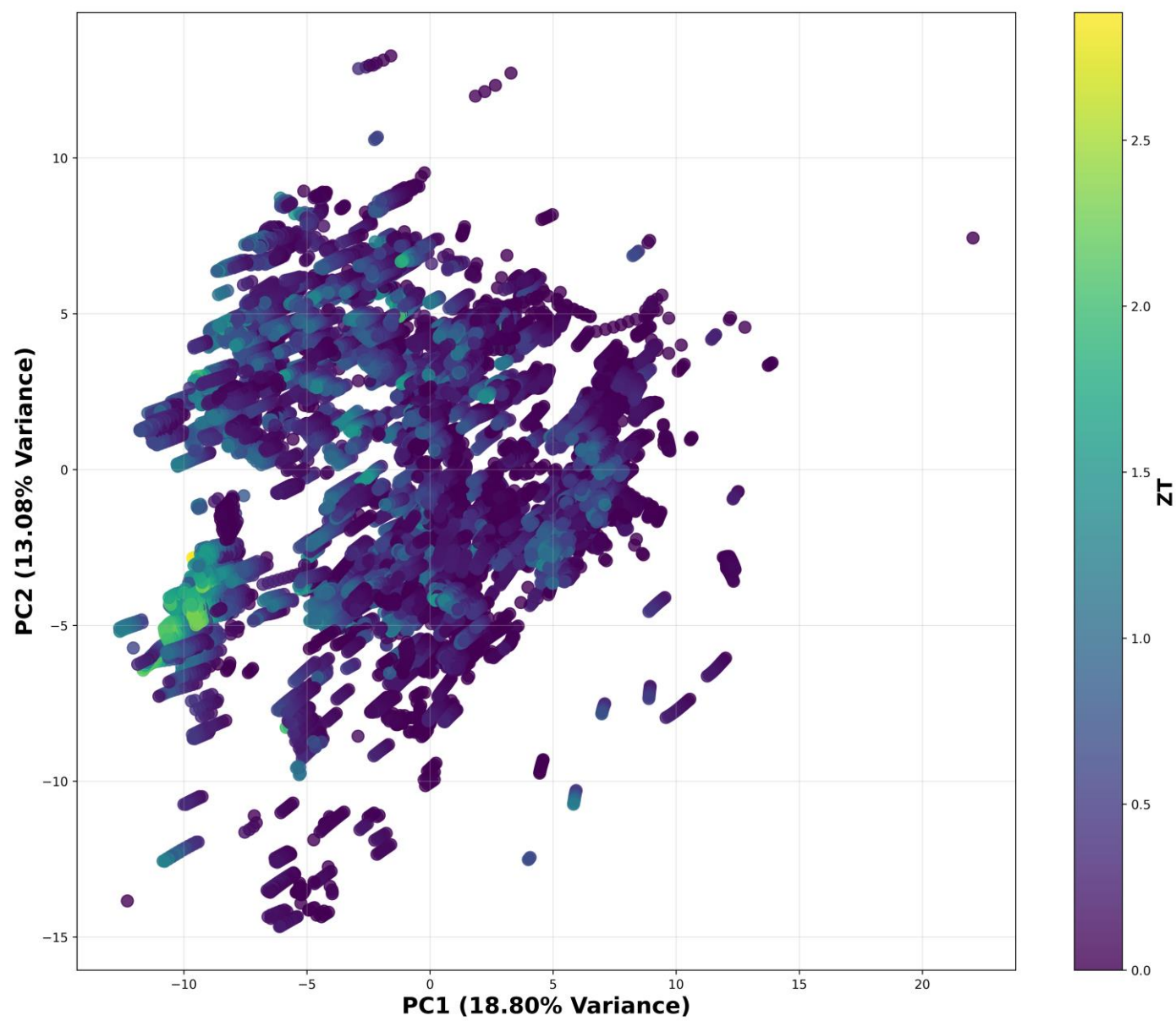
Random Forest:- 0.82

Support vector machine:- 0.85

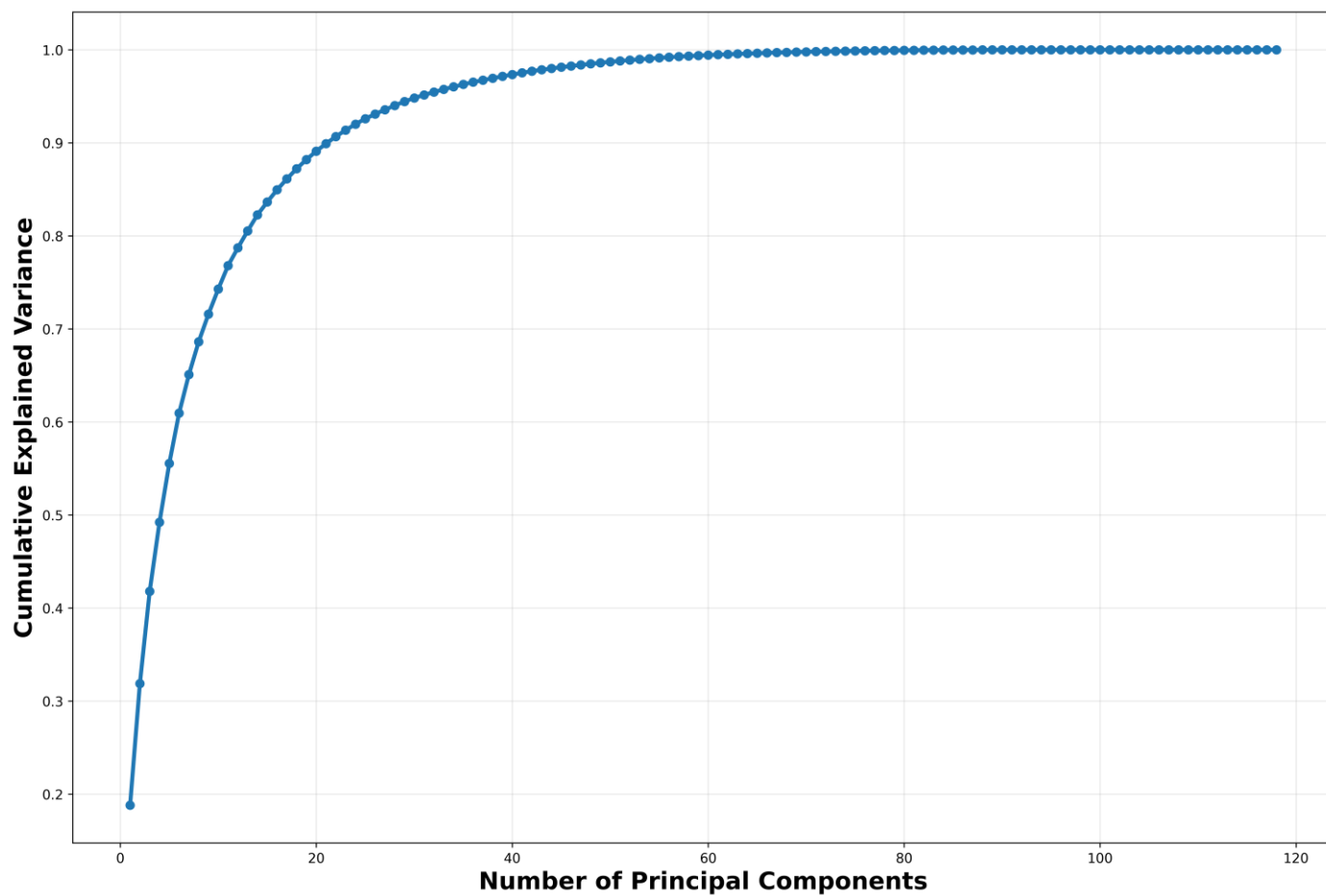
After considering the results from these models and dataset visuals we decided to go with tree based models.

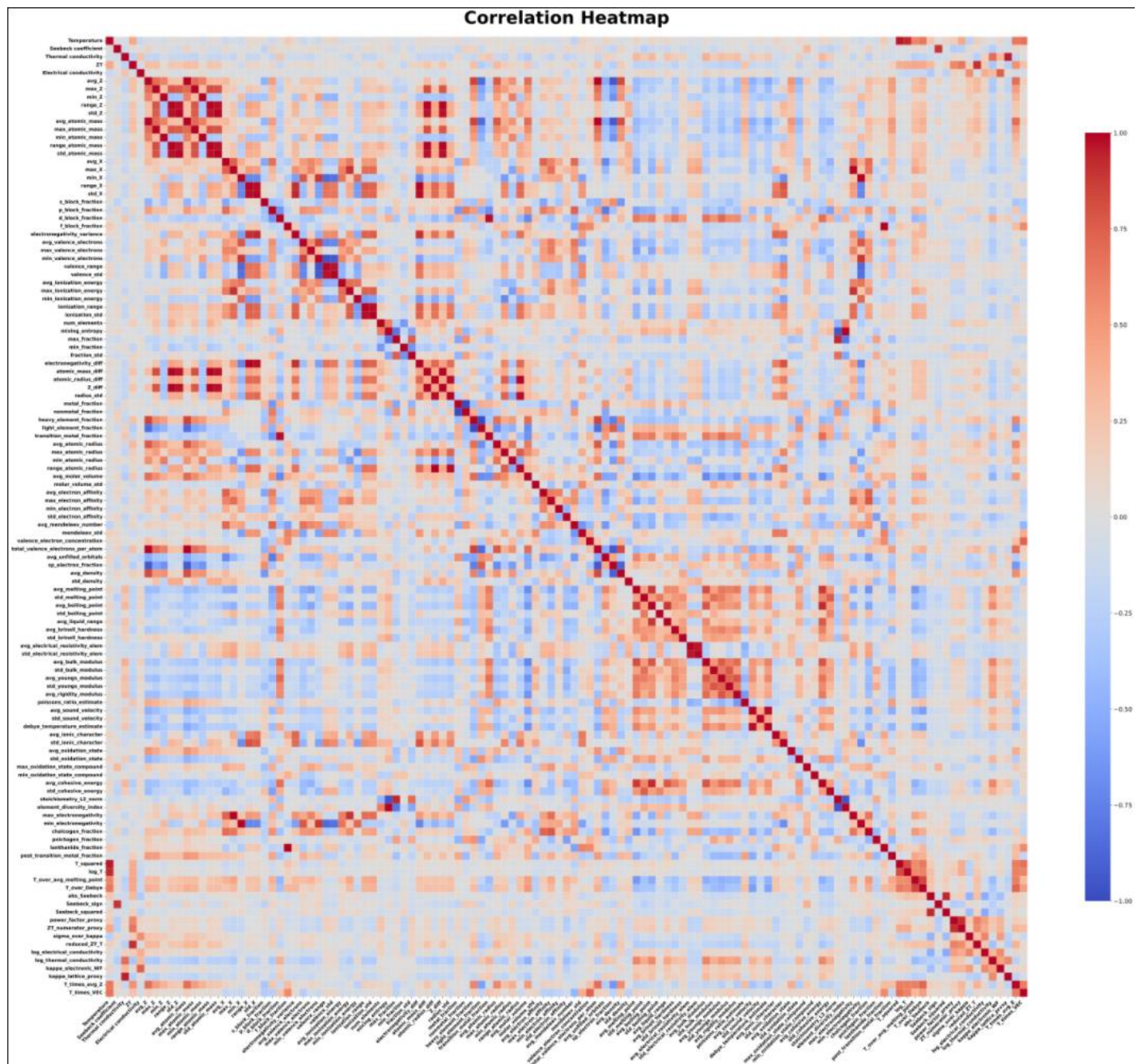
Tree-based algorithms consistently outperformed linear methods during preliminary benchmarking, indicating strong nonlinear dependencies among thermoelectric descriptors. Therefore, subsequent experiments focused on ensemble tree-learning approaches.

PCA Projection of Thermoelectric Dataset



PCA Explained Variance Curve





2. Model Development:

First to avoid data leakage , target features and data redundant features were removed from training of models

These were features that was removed from training data:

Seebeck coefficient,Thermal conductivity,Electrical conductivity,
'reduced_ZT_T','ZT_numerator_proxy','power_factor_proxy','sigma_over_kappa',
'kappa_electronic_WF','kappa_lattice_proxy', 'range_X',
'range_atomic_mass', 'range_Z', 'nonmetal_fraction','range_atomic_radius',
'total_valence_electrons_per_atom','max_electronegativity','min_electronegativity',
'light_element_fraction','element_diversity_index','lanthanide_fraction',
'max_atomic_mass','std_Z','std_X','debye_temperature_estimate','Seebeck_sign',
'log_electrical_conductivity','log_thermal_conductivity',
'T_squared','log_T', 'T_times_avg_Z','T_times_VEC',
'T_over_Debye','T_over_avg_melting_point',
'abs_Seebeck', 'Seebeck_squared'

Because some of the features were should be removed because they are part of ZT [formula] and remaining all are redundant features

CatBoost Regressor:

The CatBoost regressor was used to predict the figure of merit, ZT . It relies on compositional and physical descriptors. CatBoost is a boosting algorithm that builds decision trees sequentially. Each new tree aims to reduce the prediction errors from the previous trees. In this work, the model was trained with 300 boosting

iterations. The learning rate was set at 0.05, and the tree depth was set at 6.

This approach helps the algorithm recognize relationships in the data on thermoelectric materials. Before training, we removed features that could lead to answer leakage, redundant descriptors, and columns with numeric data. This ensures the model is scientifically sound and generalizes well. We replaced values in the dataset with the median value. The data was split into 80% for training and 20% for testing. We also used five-fold cross-validation to assess the model's robustness and reduce bias. CatBoost was chosen because it manages data and intricate interactions between descriptors effectively. It does not need feature scaling. The model achieved an R^2 score of 0.81, demonstrating its predictive ability for estimating ZT. The analysis of feature importance indicated that temperature and electronic structure descriptors are the most significant.

The Extra Trees Regressor:

The Extra Trees Regressor was used to predict ZT because it is very good at modeling complicated relationships in materials data.

For this study we trained the Extra Trees Regressor model with 200 decision trees. We used square-root feature sampling. Let the trees grow as deep as they needed to capture all the complicated interactions between the features. Before we started training we

cleaned up the data a lot. We removed features that were not helpful and got rid of duplicates. If there were missing numbers we filled them in with the value. Then we split the data into two groups one for training and one for testing.

We used a method called five-fold cross-validation to see if the model was stable and worked well with groups of data. The Extra Trees Regressor model did well with an R^2 score of about 0.8873. This made it one of the models we tried. When we looked at which features were most important we found that temperature, electronic descriptors, energy and atomic-level properties were the most important for predicting thermoelectric performance. The Extra Trees Regressor showed us that these features are very important, for predicting how well a material will work.

Gradient Boosting Machine:

The Gradient Boosting Machine regressor was used to predict the ZT values from the material descriptors we got from experiments.

In this work we set up the Gradient Boosting Machine model with 200 estimators, a learning rate of 0.1, a depth of 5 and a subsampling ratio of 0.8. We did this to make sure the model could learn well without overfitting. Before we trained the model we removed any features that were not numbers any descriptors and any variables that were directly related to ZT. This was done to make sure our predictions were reliable. We also filled in any values with the median value and we split our data into two parts: one for

training and one for testing. We used 80 percent of the data for training and 20 percent for testing.

We used a method called five-fold cross-validation to check how consistent and robust our model was. The Gradient Boosting Machine regressor was able to predict the performance very well with an R^2 score of about 0.8737. This means it is very good at predicting ZT values. When we looked at which features were most important we found that temperature, cohesive energy, valence electron concentration and electronic affinity descriptors were the important ones, for determining the predicted ZT values of the Gradient Boosting Machine.

LightGBM Regressor:

The LightGBM regressor was used for predicting ZT because it is very fast and works well with structured materials datasets. LightGBM is a tool that builds trees in a special way, which helps it learn faster and make better predictions. In this project the LightGBM model was trained with 300 boosting estimators, a learning rate of 0.05 and a maximum tree depth of 6 with 63 leaves per tree. This helped the model learn the relationships between the descriptors that affect thermoelectric behavior. The dataset was cleaned up by removing features that were not relevant, duplicates and non-numeric descriptors to make sure the results were reliable. Missing numbers were filled in with the value and the dataset was split into training and testing sets. The model was tested times to see how well it worked with different parts of the data.

LightGBM is good at handling datasets with many complex relationships and it does not need feature scaling. with an R^2 score of 0.8875 which is one of the best results. The important features that affect thermoelectric ZT prediction are temperature, electronic affinity, cohesive energy, oxidation state and valence electron concentration as shown by the feature importance analysis of the LightGBM regressor. The LightGBM regressor performed well in this project.

XGBoost Regressor:

The XGBoost regressor was used to predict ZT values because it has a strong framework and can control itself well.

In this study we trained the model with 300 estimators, a learning rate of 0.05 and a maximum tree depth of 6. We also used strategies to select rows and features to make the model work better with data. Before training the model we cleaned the data by removing information redundant variables and non-numeric columns. We then filled in missing values with the value.

We split the cleaned data into two sets: one for training and one for testing. The training set was 80 percent of the data and the testing set was 20 percent. We also did five- cross-validation to check if the model was stable and consistent. The XGBoost regressor did well with an R^2 score of about 0.8857. This means it is very good at modeling how thermoelectric materials behave.

When we looked at what features were important we found that valence range, electron fraction, cohesive energy, atomic descriptors and temperature had an influence on the predicted ZT values. This shows that the XGBoost regressor effectively captured the underlying physics of materials. The XGBoost regressor is a tool, for predicting thermoelectric ZT values.

Deep Learning Model (ZTNet – Residual MLP):

The Deep Learning Model is called ZTNet. The MLP model functions as the main structure of ZTNet. The ZTNet learning framework implements a Residual MLP architecture for its learning processes. The ZTNet system predicts the measurement known as ZT by analyzing material descriptors obtained through experimental methods. The ZTNet system uses deep neural networks for data learning purposes which distinguishes it from other machine learning systems. The deep neural networks in this system connect their hidden layers to each other. The system learns about the data through a specific method. The developers needed to clean the data before they could achieve optimal results with ZTNet. The process required them to eliminate specific variables which did not contribute to the analysis. The system used non-numeric features to create operations between those features. The system replaced missing numbers with the value of existing numbers. The system standardized all input features to enhance neural network performance. The data was divided into three segments which included a training section and a validation section and a testing section. The ZTNet performance assessment took place during its

training phase. The ZTNet system has an architectural design which begins with a data input layer. The system combines four interconnected blocks with its operational design. The blocks include batch normalization and ReLU activation together with dropout regularization and skip connections that enhance network performance. The ZTNet model used AdamW optimizer together with cosine annealing learning rate scheduler for its training process. The training process continued for 100 complete cycles. The system used mean squared error to determine its loss value. The ZTNet system learned how material descriptors connect to thermoelectric performance through its learning process. The system avoided overfitting because it did not acquire knowledge about nonexistent data patterns. The ZTNet system achieved high ZT prediction accuracy through its R^2 score of 0.8745.

It had prediction error, which means it was very accurate. The training and validation loss curves showed that the ZTNet was working well and was not overfitting. The ZTNet is a Deep Learning Model that's very good at modeling thermoelectric materials. The ZTNet is based on a MLP architecture and it is very good at predicting the ZT.

3. Integration of models:

We have now integrated top 2 models out of 8 models we trained above , after comparing LightGBM have the most accuracy and after that we have ExtraTrees and after that we have XGBoost and after that GBM .

Instead of selecting a single best-performing algorithm, top-performing models were integrated through stacked ensemble

learning to exploit complementary prediction behaviour and improve generalization.

Now we integrated them as one now we have 2 integrative models

1. LightGBM-ExtraTrees.

2. XGBoost-GBM

ExtraTrees + LightGBM Stacked Ensemble Model:

The researchers created an ensemble learning system which combined Extra Trees and LightGBM regression models to achieve better results in thermoelectric ZT predictions through the use of multiple machine learning methods that work together. The prediction results from multiple models in ensemble learning create better generalization results which increase system stability when compared to predictions made by single models. The research employed Extra Trees regressor to detect all random nonlinear interactions between different descriptor variables while LightGBM regressor provided efficient modeling of complex gradient-boosted patterns in the thermoelectric dataset. The Ridge regression meta-learner within the stacking architecture combined outputs from both base learners to create the final prediction result. The research team conducted extensive data cleaning activities before training which included the elimination of target leakage descriptors and redundant variables and non-numeric features together with median imputation for handling missing data. The dataset was divided into training and testing sets using an 80:20 split ratio and

five-fold cross-validation was used to assess how well the model performed while preventing overfitting. The stacked ensemble achieved an R^2 score of approximately 0.920 which represented the highest predictive performance among all investigated models.

The prediction errors showed a normal distribution pattern since they concentrated around zero which confirmed the model's strong stability and minimal systematic bias. The analysis of feature importance showed that temperature functioned as the primary factor which determined thermoelectric performance while cohesive energy and valence-related descriptors and electron affinity and atomic-scale compositional features followed temperature in importance. The ensemble model achieves better results because it combines multiple nonlinear learners which improve the ability to forecast complex thermoelectric behavior.

Model reliability :

To reduce the possibility of information leakage, model preprocessing included removal of descriptors directly derived from ZT-related quantities and Leakage-sensitive preprocessing was fitted only using training folds. Test data remained unseen. prior to evaluation. Five-fold cross-validation was used to estimate generalization stability.

XGBoost + Gradient Boosting Stacked Ensemble

Model:

The stacked ensemble model which uses both XGBoost and Gradient Boosting Machine (GBM) regressors achieves high-accuracy results for thermoelectric ZT prediction. Stacked ensemble learning improves predictive capability by combining multiple machine learning models with different learning behaviors into a unified framework. The XGBoost regressor was used in this research

because it provides strong regularization and optimized gradient boosting capabilities while the GBM model delivered effective residual error correction and nonlinear feature learning. The final ensemble prediction was generated by combining predictions from

two base learners through a Ridge regression meta-learner which operated within a stacking architecture.

Before starting the model training process, researchers conducted multiple preprocessing operations which included deleting all features that contained target leakage and duplicate descriptors and non-numeric variables, and they handled missing data through median imputation. The researchers divided the cleaned dataset into training and testing subsets which they split using an 80:20 ratio, while they used five-fold cross-validation to evaluate both generalization performance and model stability. The ensemble model achieved an R^2 score of approximately 0.901 which demonstrates its ability to accurately predict thermoelectric material properties. The prediction error analysis showed that prediction errors stayed close to zero while only showing slight variations which proved that the model operated consistently with minimal systematic errors. The analysis of feature importance showed that temperature, cohesive energy, valence range, electron fraction, density, and atomic descriptors functioned as the key factors that determined thermoelectric performance. the modeling of complex nonlinear relationships present in thermoelectric materials datasets.

4. Similarity-Guided Inverse Design:

Inverse Design and Materials Generation:

The trained ExtraTrees and LightGBM stacked ensemble model was used as a surrogate predictor inside a similarity-guided inverse design framework. that found thermoelectric material candidates with desirable physical properties and high ZT value predictions. The entire inverse design process included seed material selection, similarity-guided perturbation, surrogate prediction, Tanimoto similarity filtering, and physical sanity checks.

Engineers chose seed materials that were the best thermoelectric options from the experimental dataset. The high-ZT compounds served as fixed starting points for researchers to explore the compositional feature space. They applied controlled Gaussian perturbations to generate multiple candidate materials from each seed composition within the normalized feature domain.

The generated candidates were assessed using the trained stacked ensemble surrogate model. Their similarity to the original seed materials was measured using the continuous Tanimoto similarity metric. The system enforced a similarity threshold that required new designs to maintain both chemical and physical similarities with materials verified by existing experimental data. Only materials with predicted ZT

values exceeding the standard for thermoelectric performance were allowed to advance.

The inverse design stage should be interpreted as constrained candidate exploration rather than direct material discovery. Because seed materials were selected from the highest-performing experimentally reported compounds, generated candidates were not expected to systematically exceed seed ZT values.

Instead, the objective was to identify nearby candidate regions that preserved physical plausibility while maintaining useful thermoelectric performance.

Generated candidates exhibited high similarity to real materials (mean Tanimoto similarity ≈ 0.99), indicating that perturbations remained inside experimentally supported descriptor space. The highest predicted candidate achieved $ZT \approx 1.51$ while satisfying similarity and physical consistency constraints.

These results suggest that the proposed workflow can support hypothesis generation and prioritization of candidate materials for future computational or experimental validation.

Limitations of Inverse Design

The generated candidates are surrogate-model predictions and do not represent experimentally validated compounds. No density functional theory (DFT) or laboratory verification was performed. Therefore, generated candidates should be interpreted as screening outputs rather than confirmed thermoelectric discoveries.

5. SHAP-Based Model Interpretability Analysis:

SHAP analysis was performed to improve interpretability of the ensemble workflow. Tree-based SHAP analysis showed that temperature contributed the largest average impact on model output.

The SHAP dependence analysis demonstrated a generally positive contribution of increasing temperature toward predicted ZT values within the observed domain.

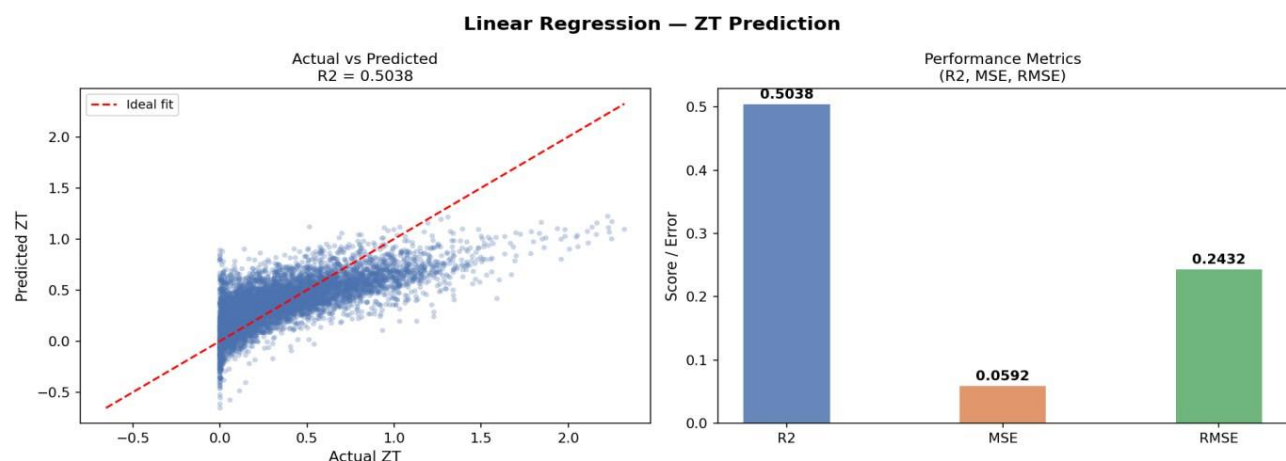
Additional influential descriptors included heavy-element fraction, average cohesive energy, electron-related descriptors, atomic mass statistics, and compositional diversity measures.

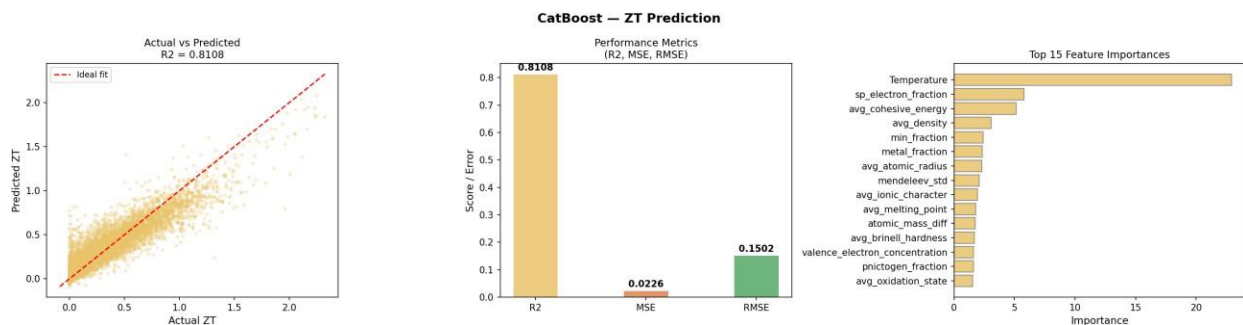
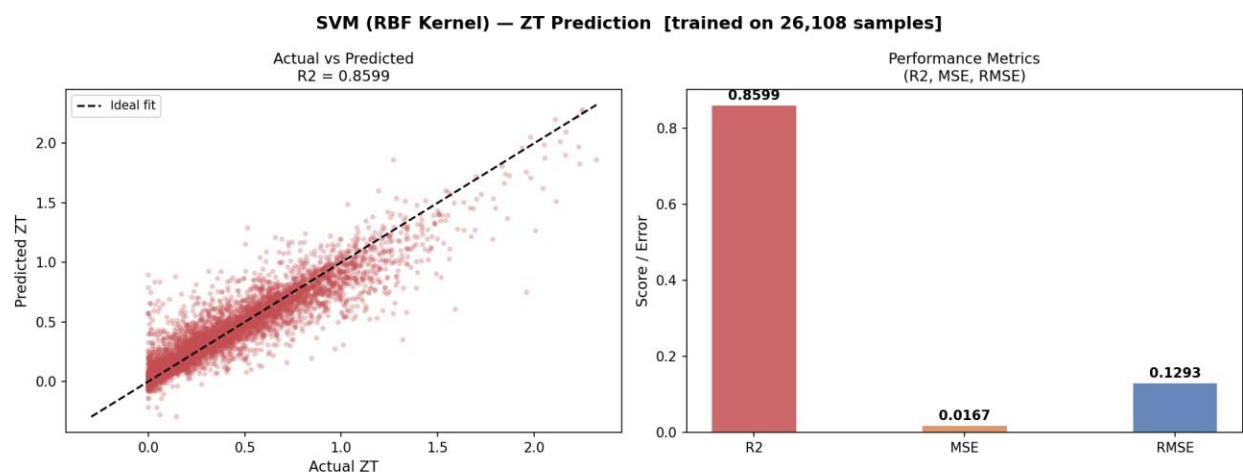
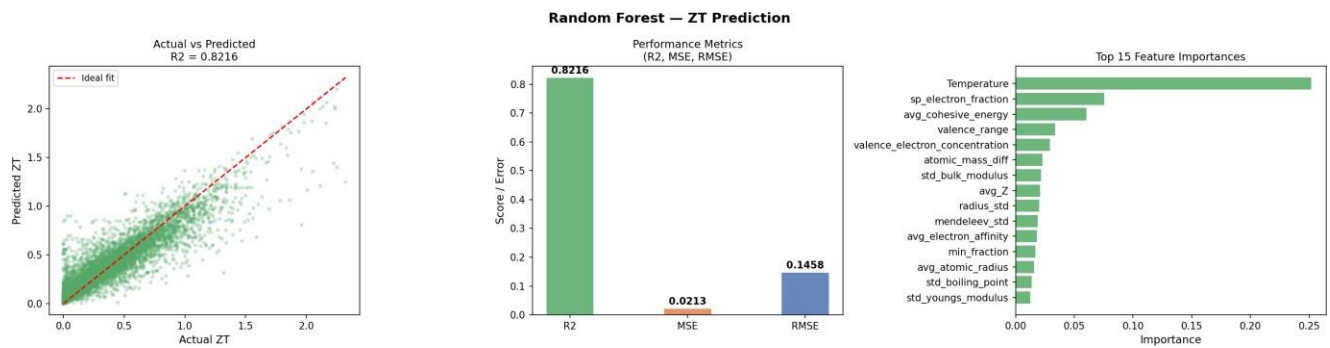
Agreement between ensemble feature importance and SHAP interpretation suggests that model decisions remain physically interpretable rather than purely statistical.

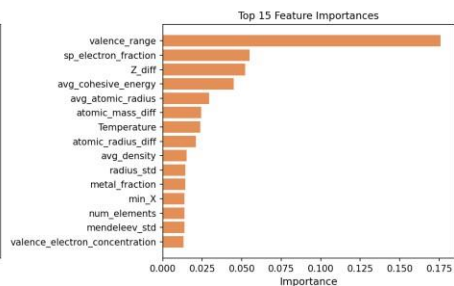
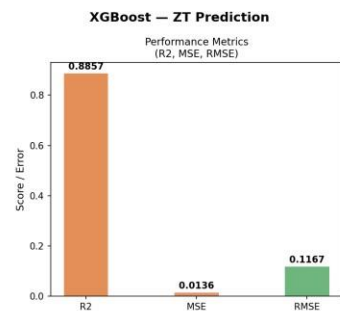
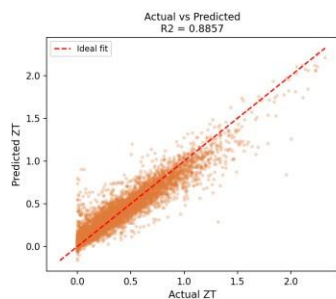
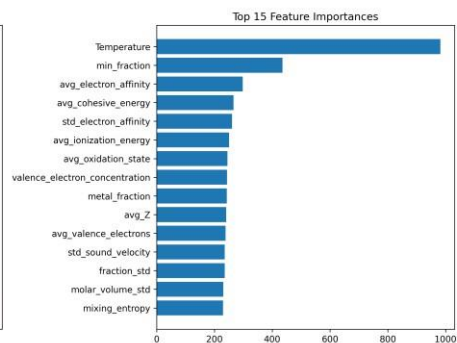
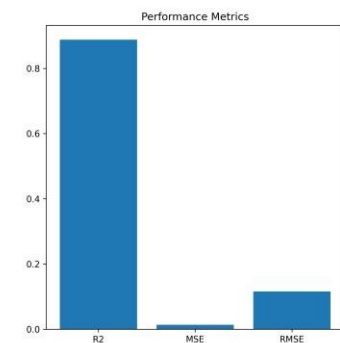
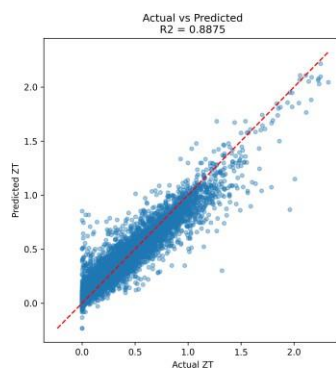
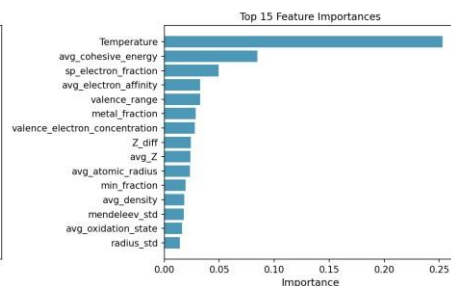
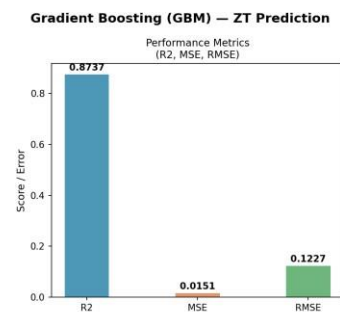
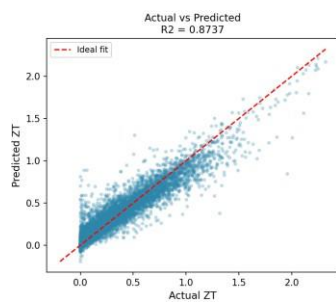
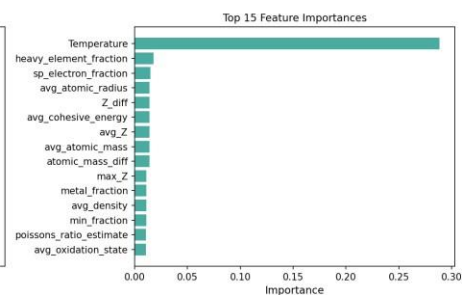
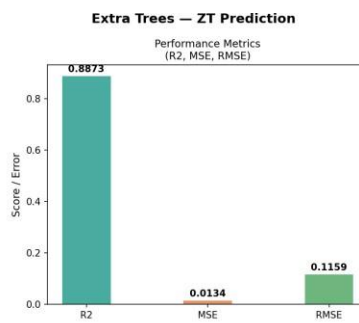
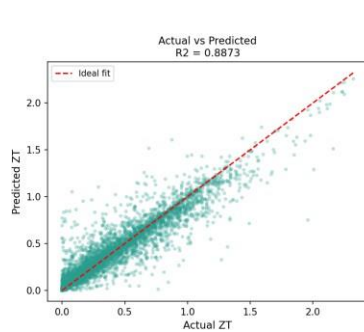
Results:

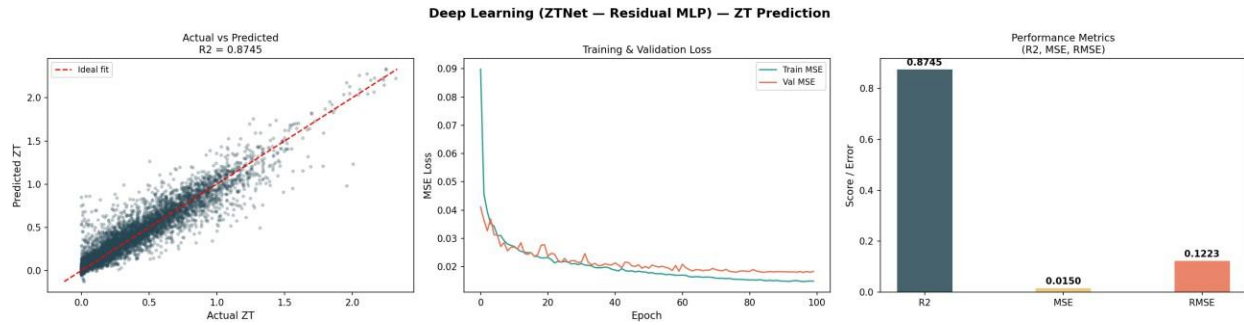
Model	Type	Test R ²	Test MSE	Test RMSE	Rank
CatBoost	Gradient Boosting Ensemble	0.8108	0.0226	0.1502	6
Extra Trees	Randomized Tree Ensemble	0.8873	0.0134	0.1159	4
Gradient Boosting Machine (GBM)	Sequential Boosting Ensemble	0.8737	0.0151	0.1227	7
LightGBM	Leaf-wise Gradient Boosting	0.8875	0.0130	0.1140	3
XGBoost	Regularized Gradient Boosting	0.8857	0.0136	0.1167	5
ZTNet (Residual MLP)	Deep Learning Neural Network	0.8745	0.0150	0.1223	8
ExtraTrees + LightGBM Ensemble	Stacked Ensemble Model	0.9202	—	0.0975	1
XGBoost + GBM Ensemble	Stacked Ensemble Model	0.9008	—	0.1090	2

Graphs and plotting of each model:



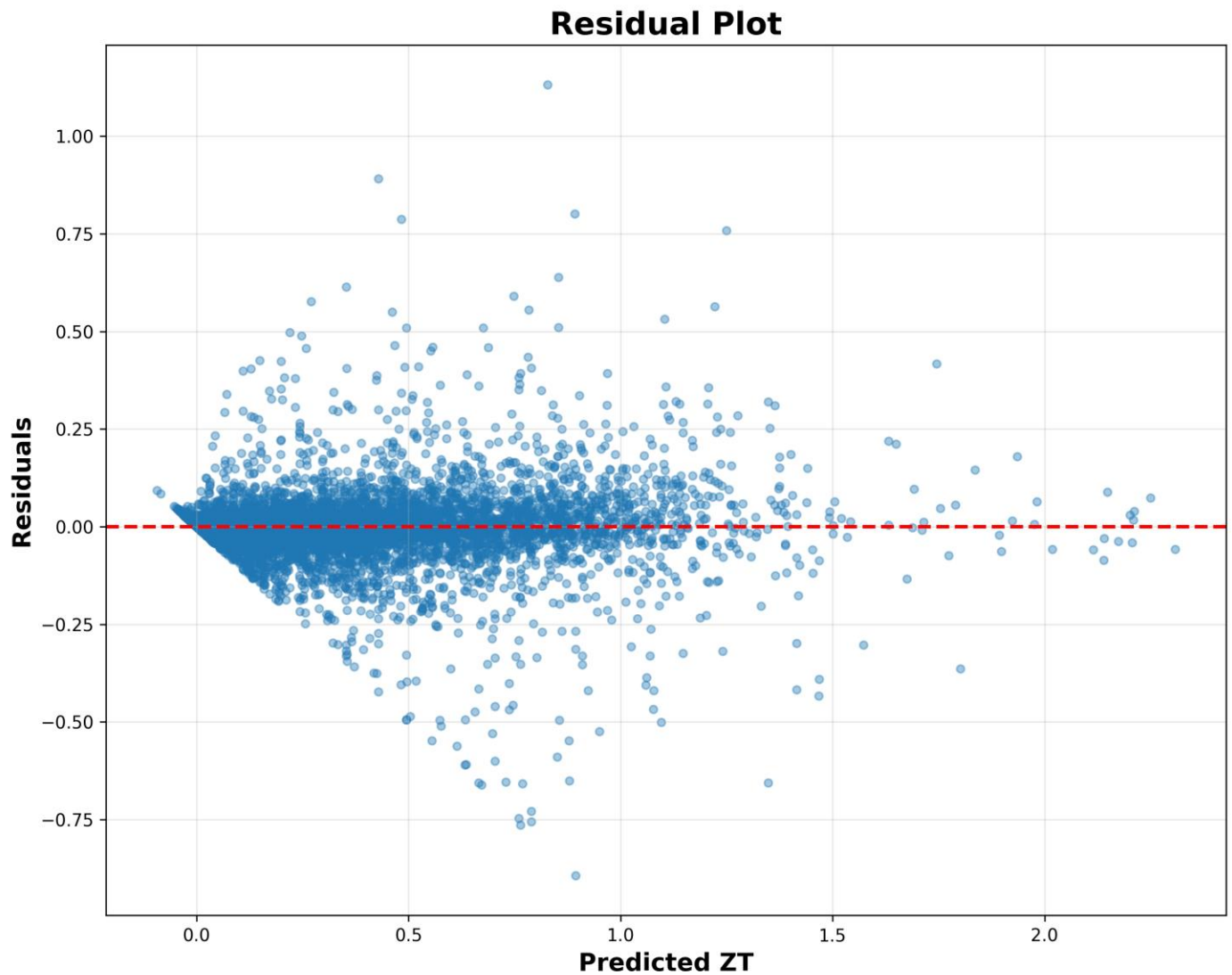




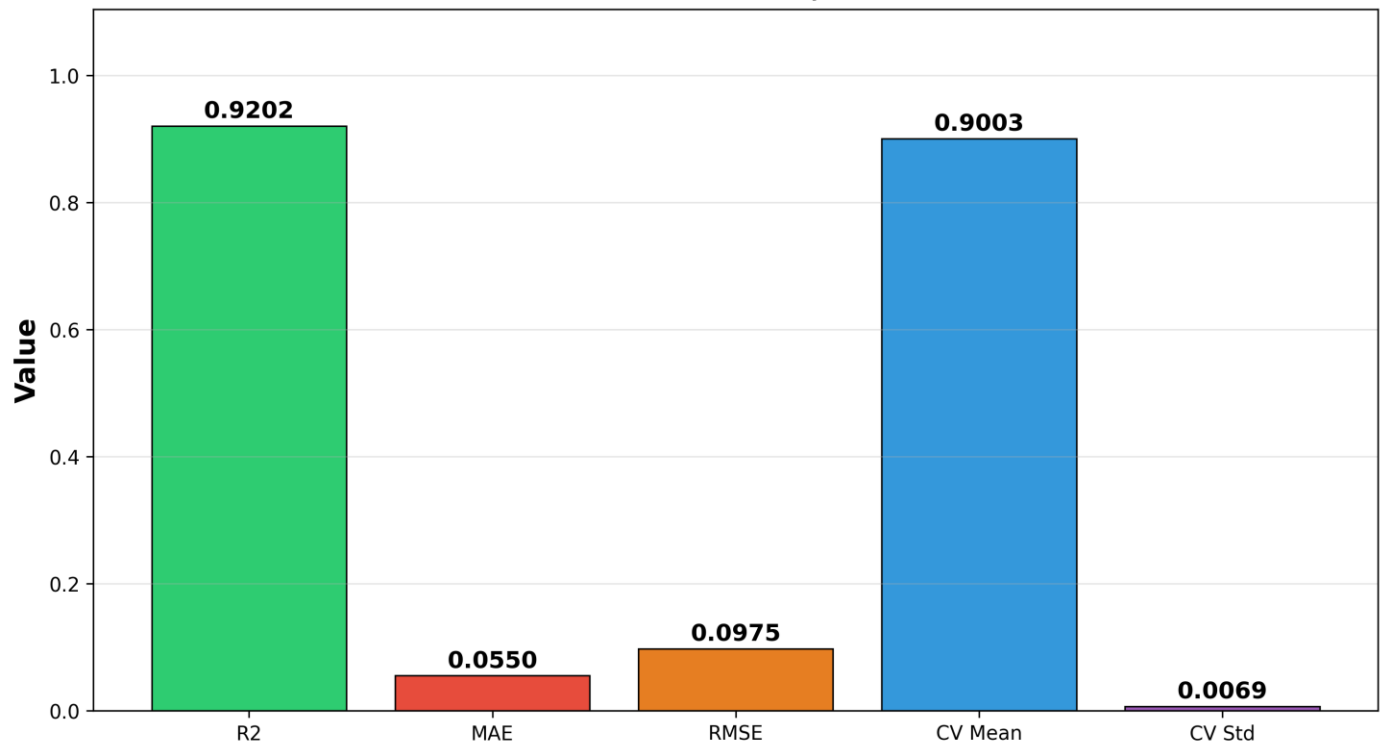


Integrative model graphs and plots:

LightGBM-ExtraTress:

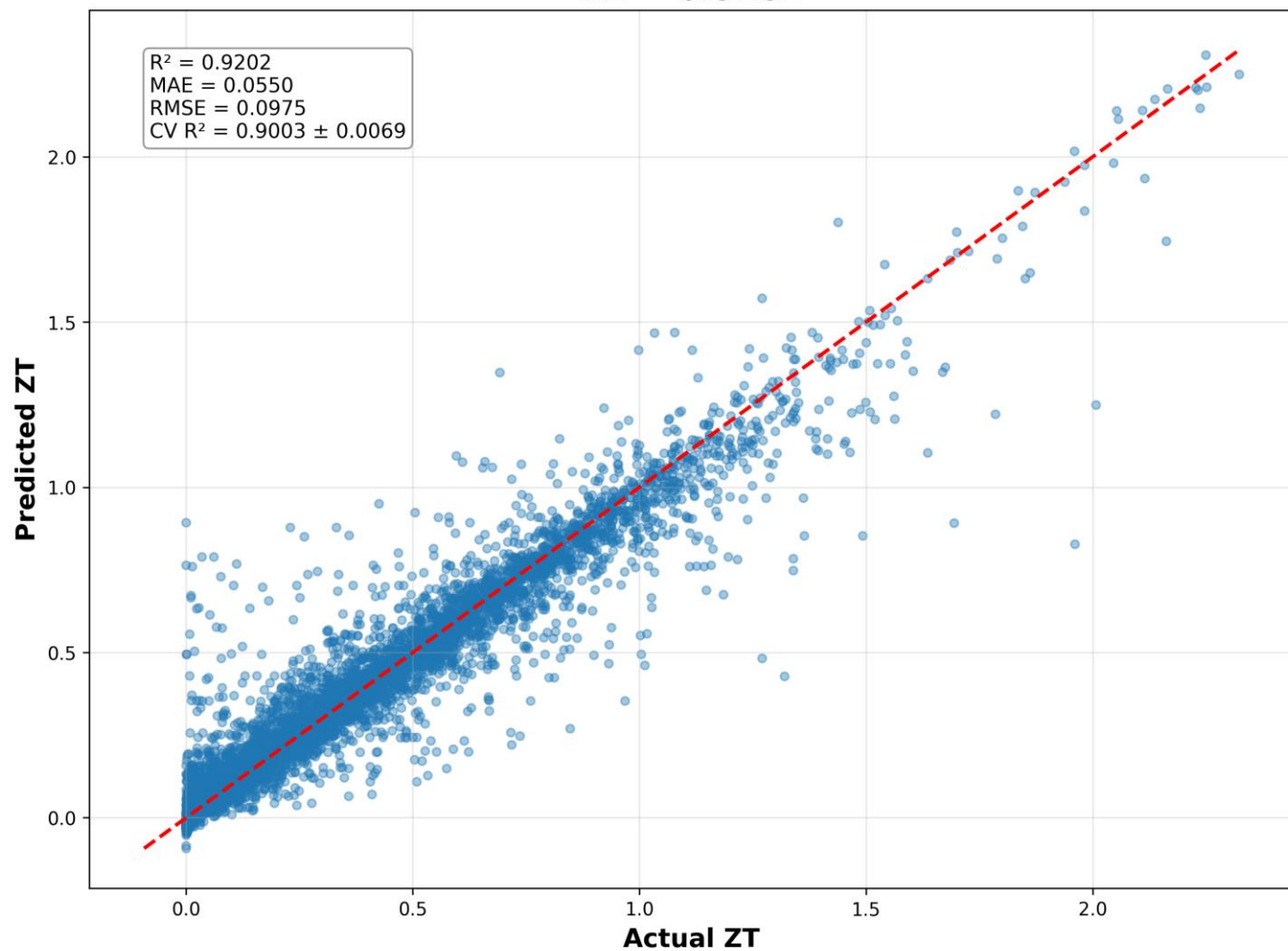


ExtraTrees + LightGBM Ensemble -- Metrics Summary
CV R2 = 0.9003 +/- 0.0069

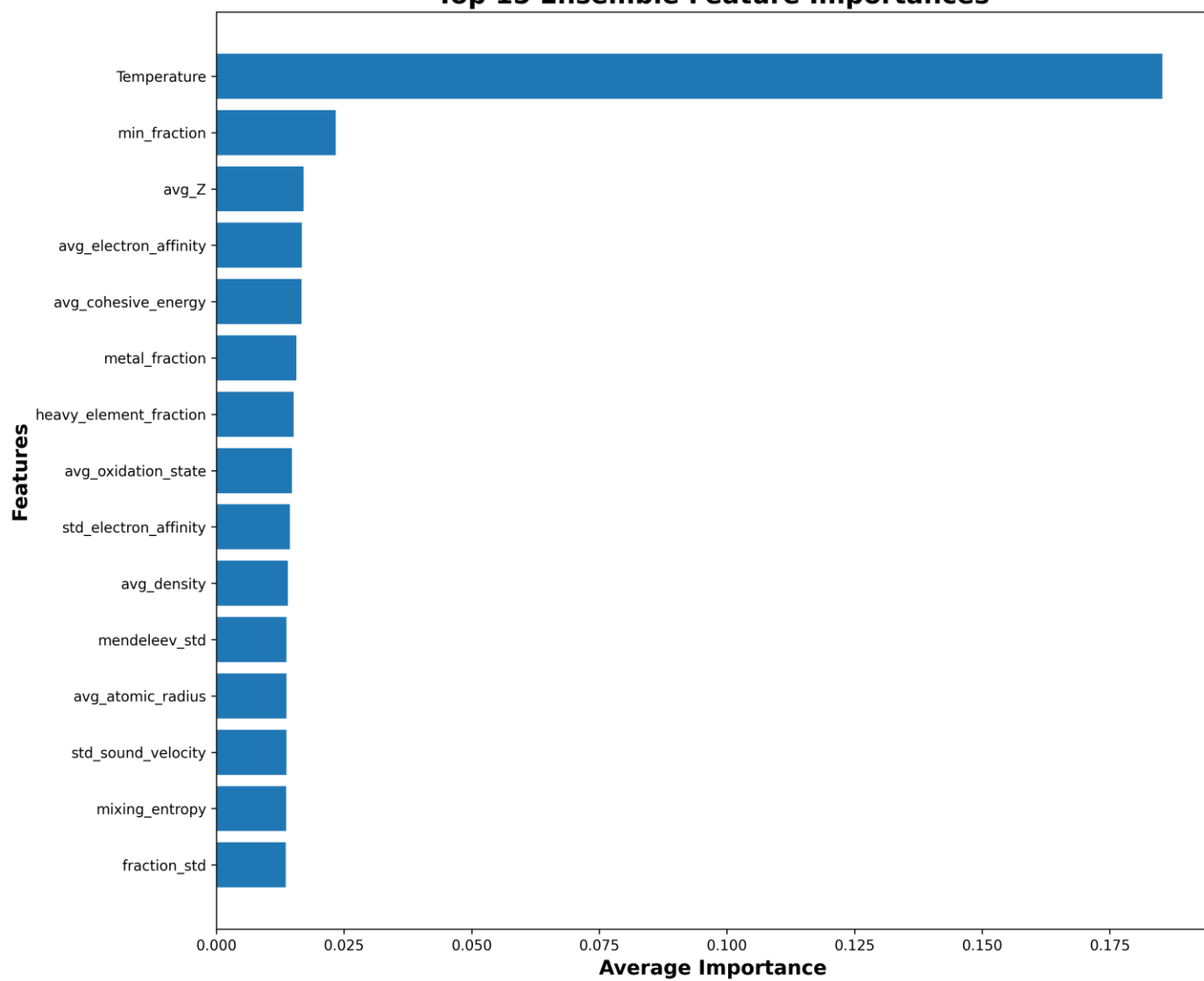


ExtraTrees + LightGBM Ensemble

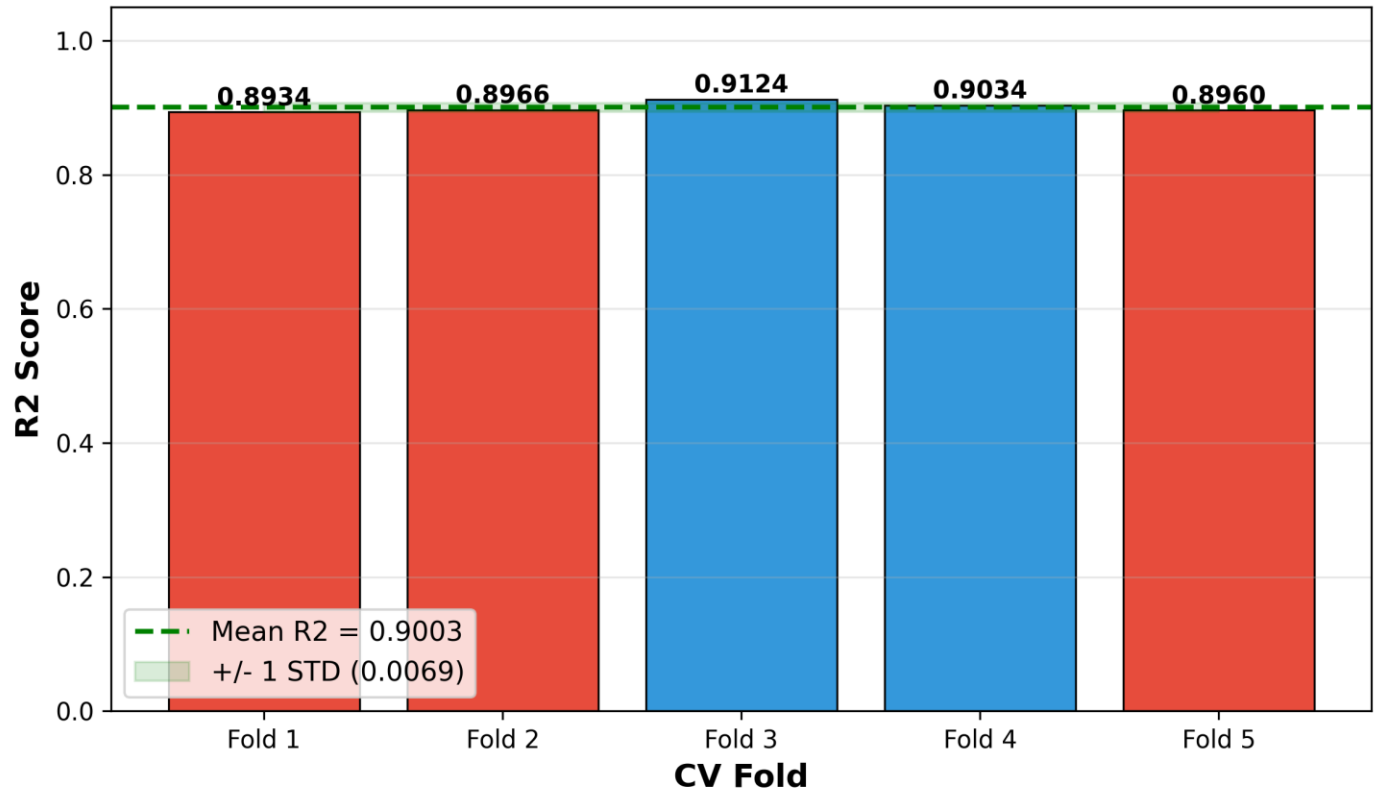
R² = 0.9202



Top 15 Ensemble Feature Importances

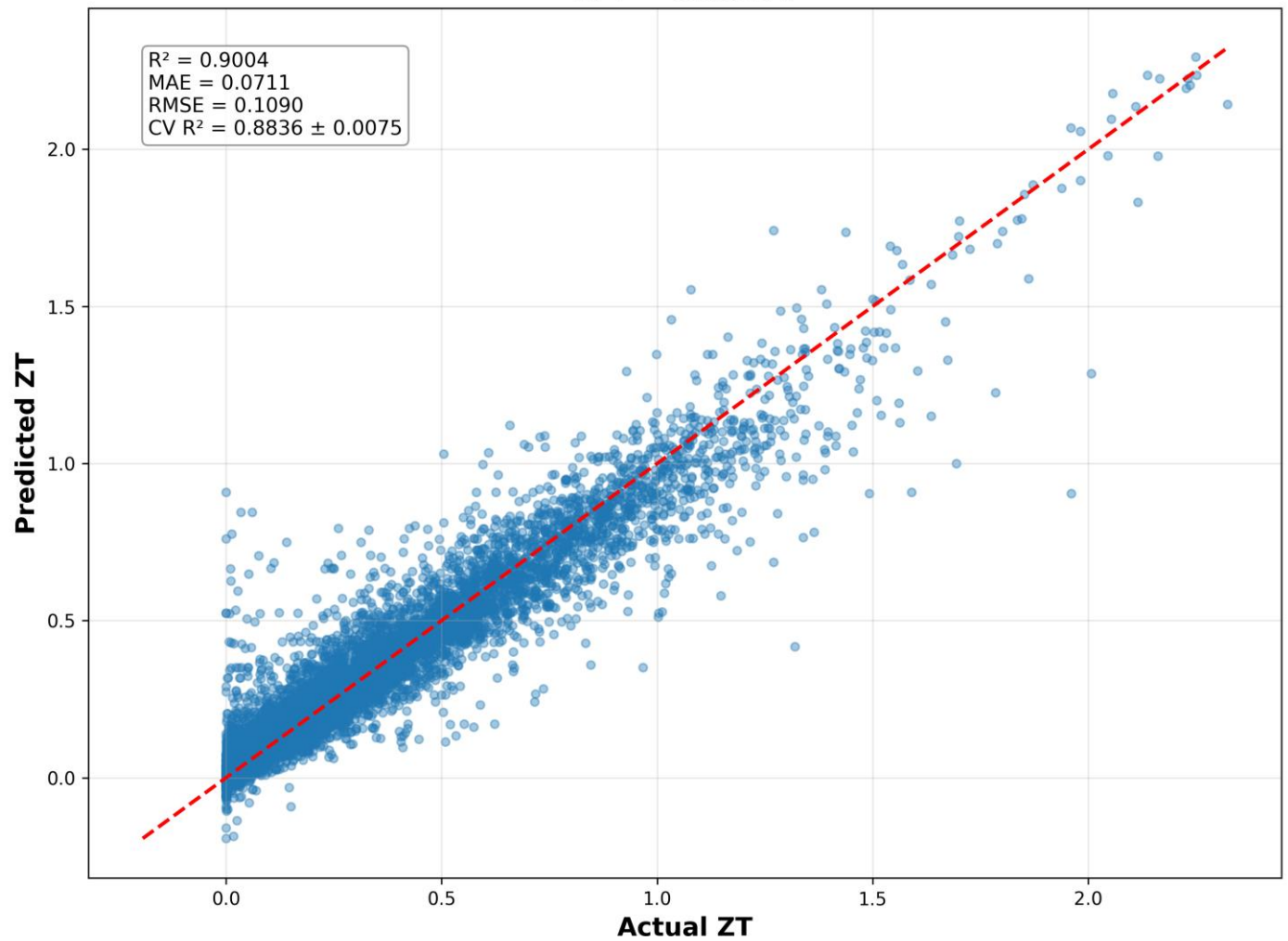


5-Fold Cross Validation R2 Scores

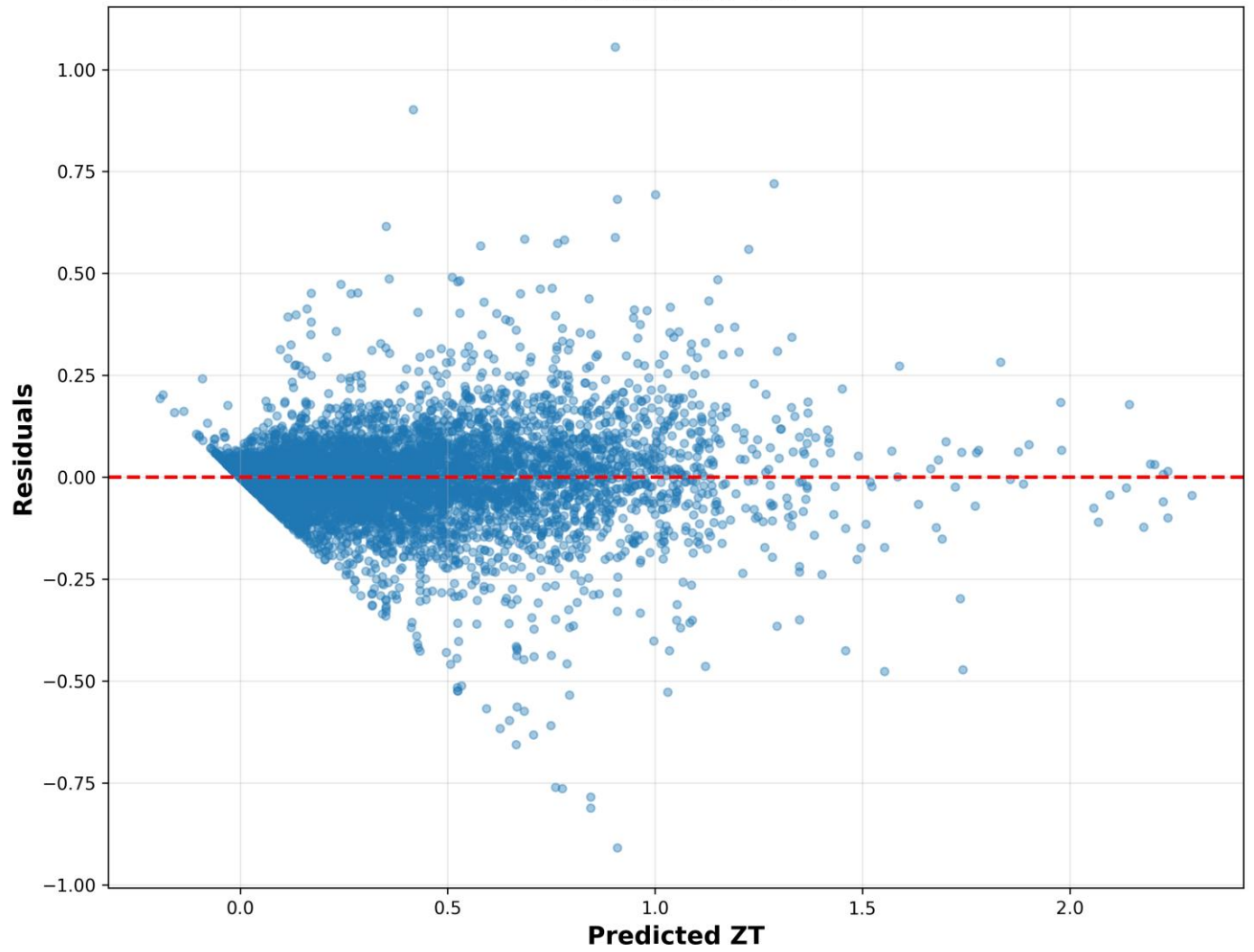


XGBoost-GBM:

XGBoost + GBM Ensemble
R2 = 0.9004

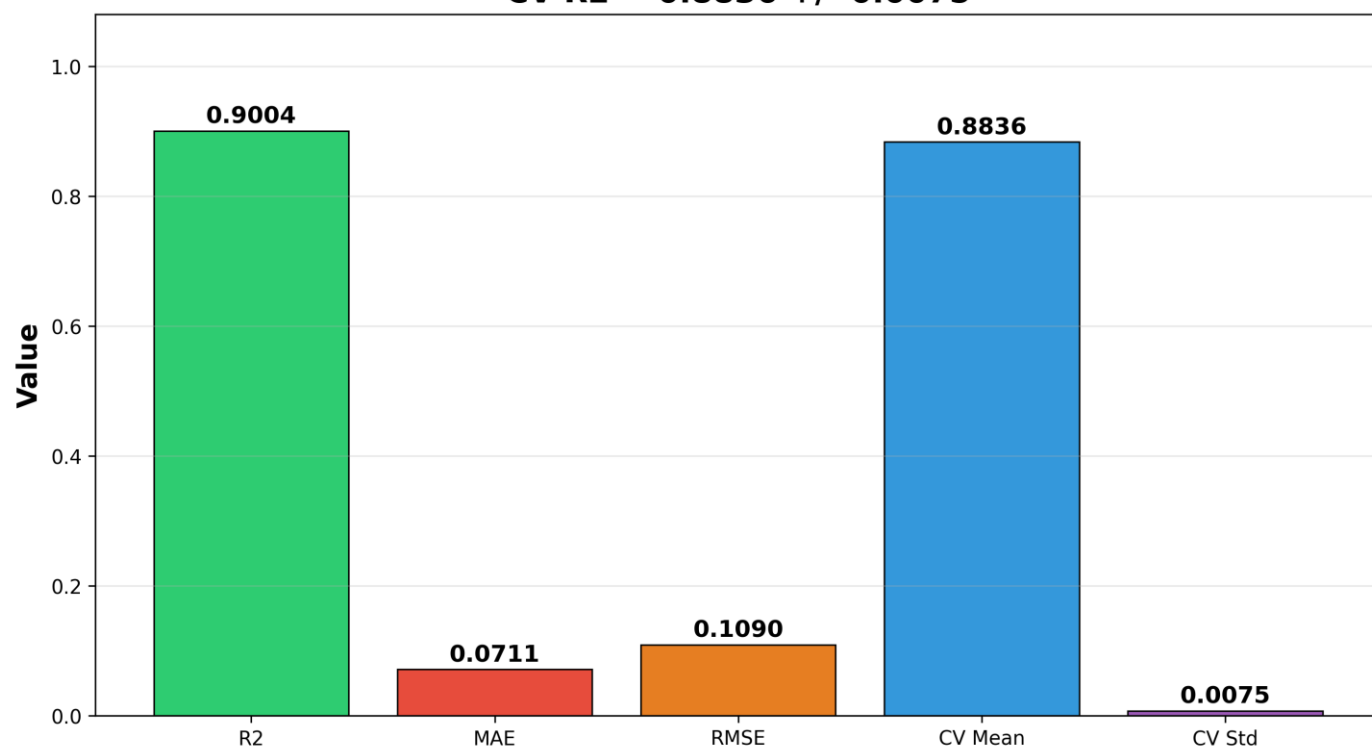


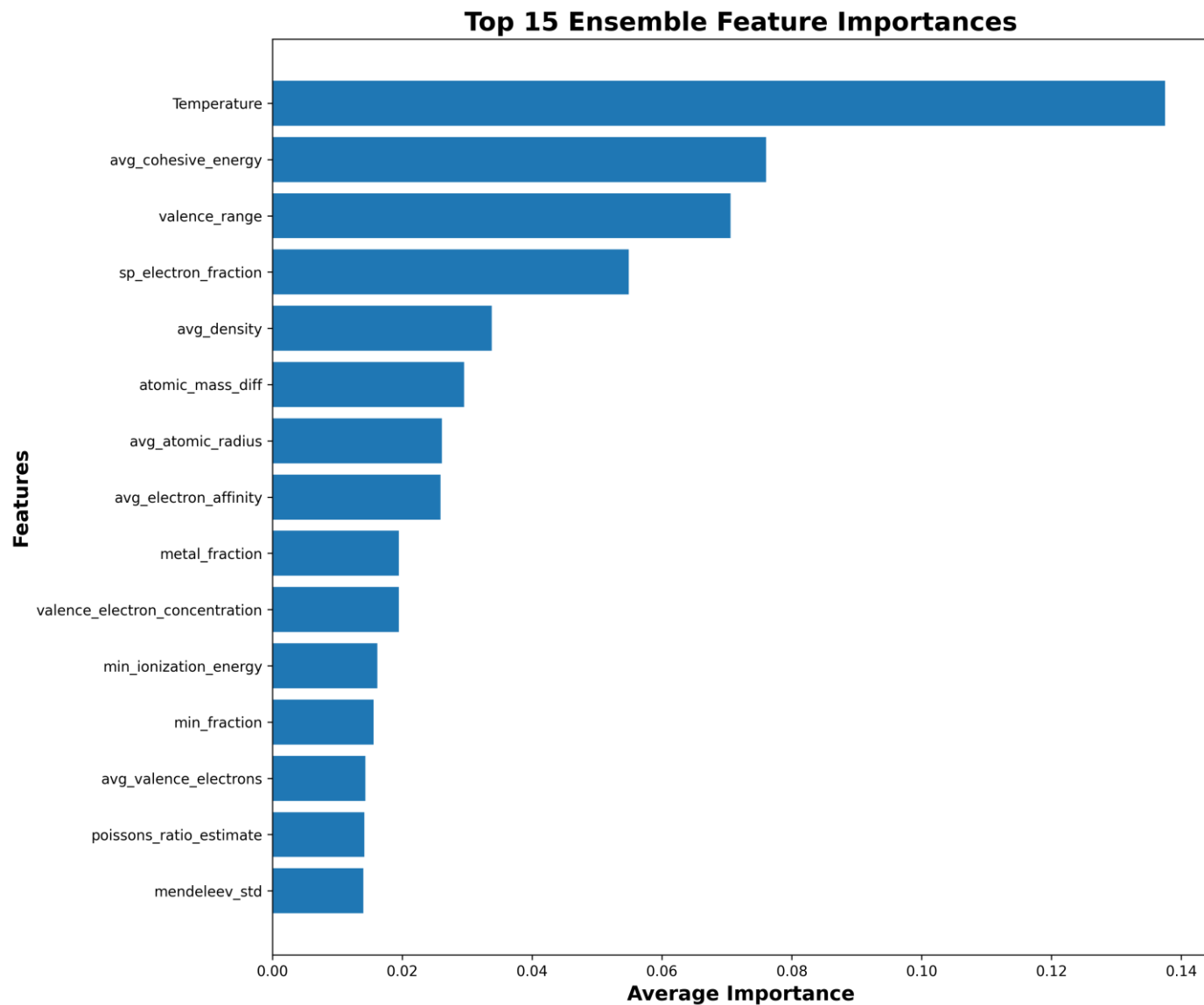
Residual Plot

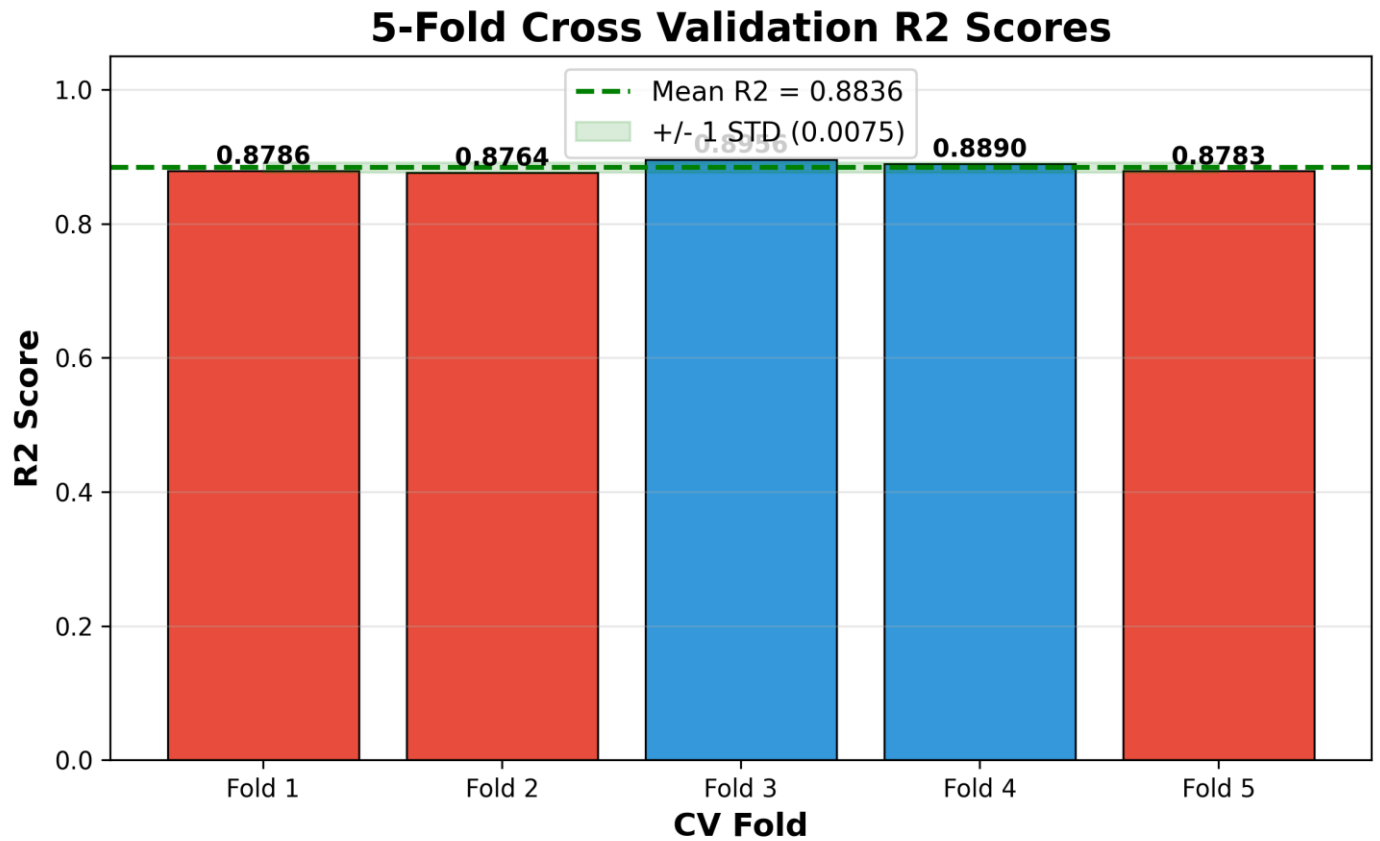


XGBoost + GBM Ensemble -- Metrics Summary

CV R2 = 0.8836 +/- 0.0075

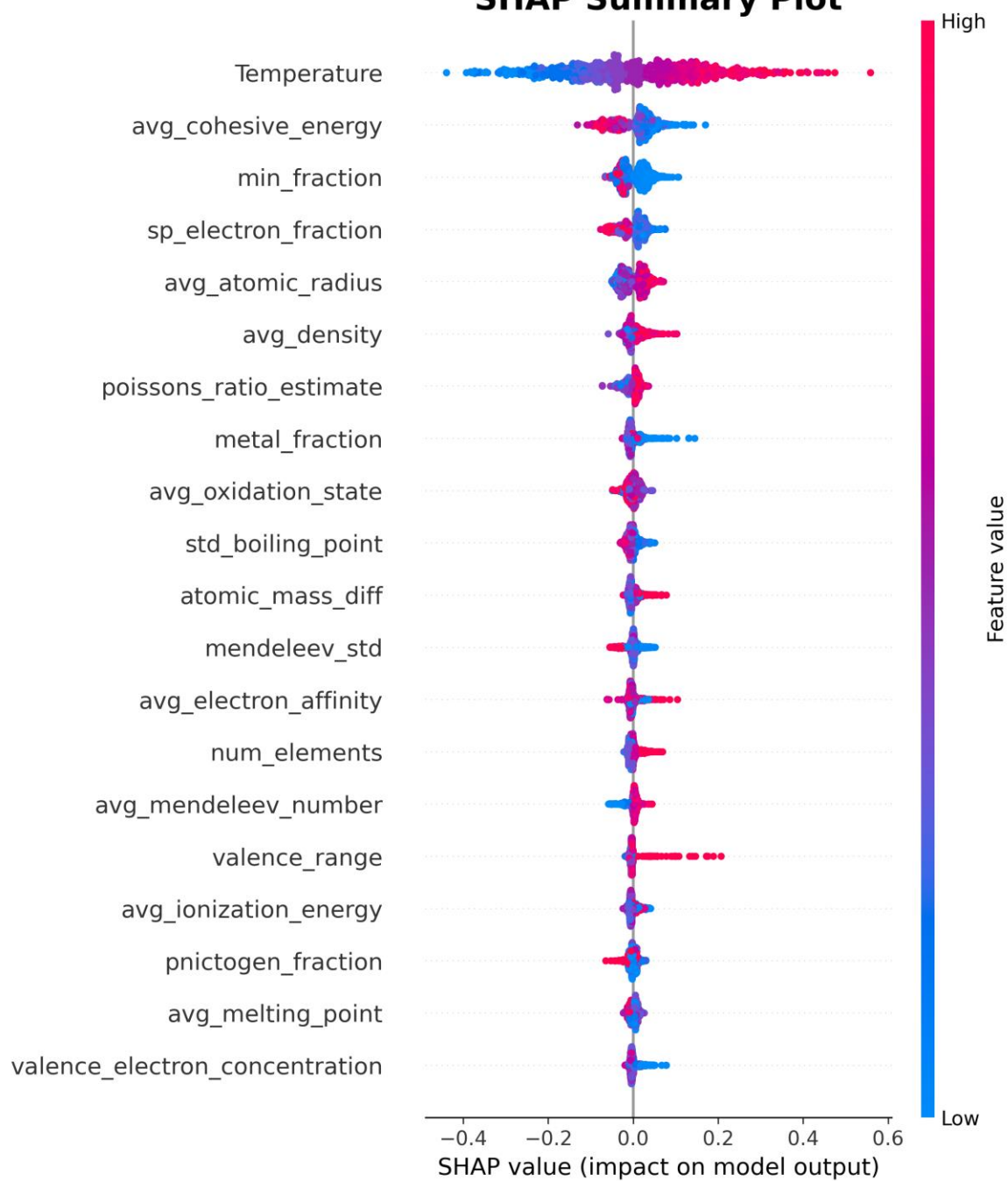




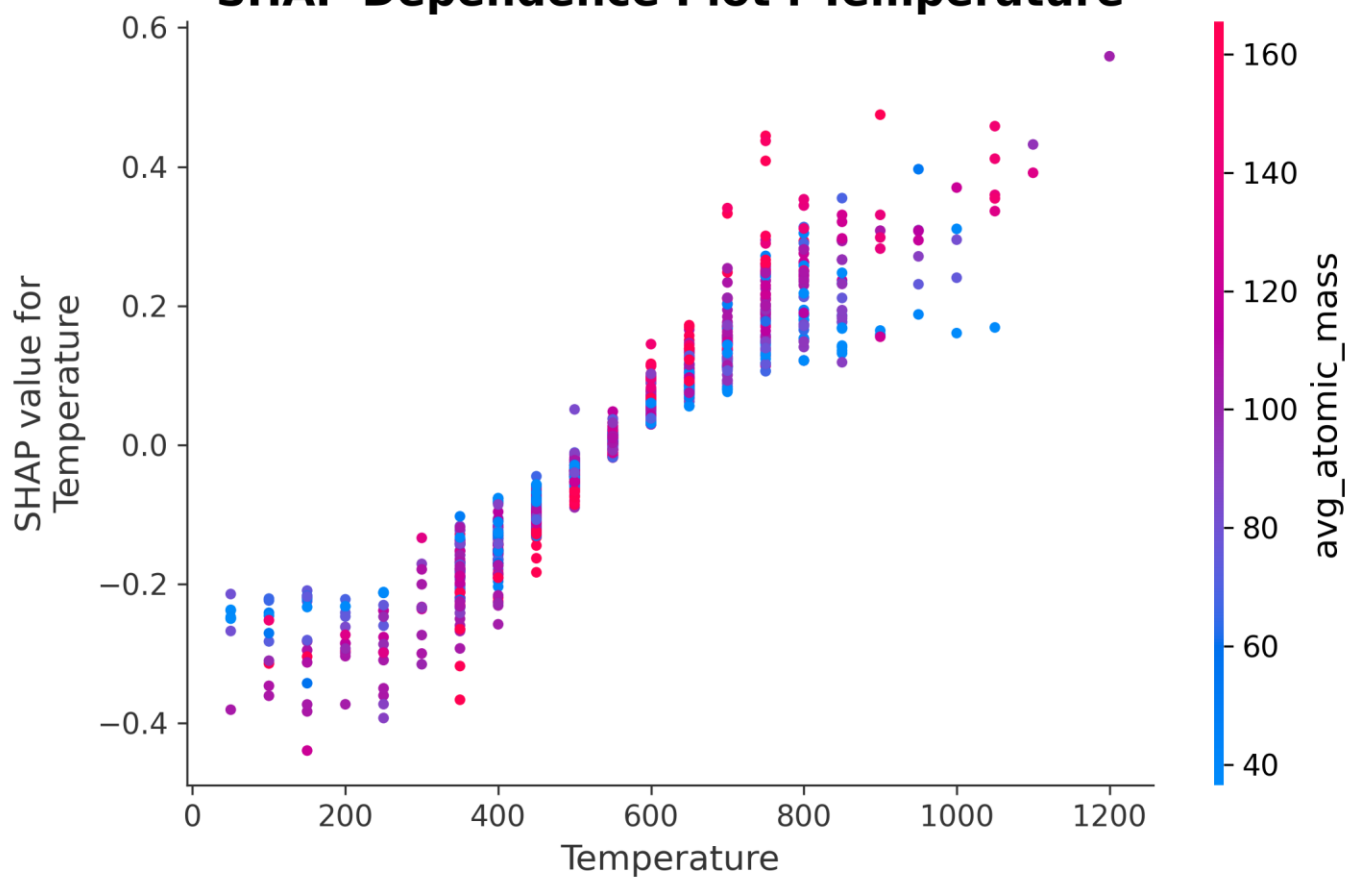


Graph of SHAP Analysis for the LightGBM-ExtraTress:

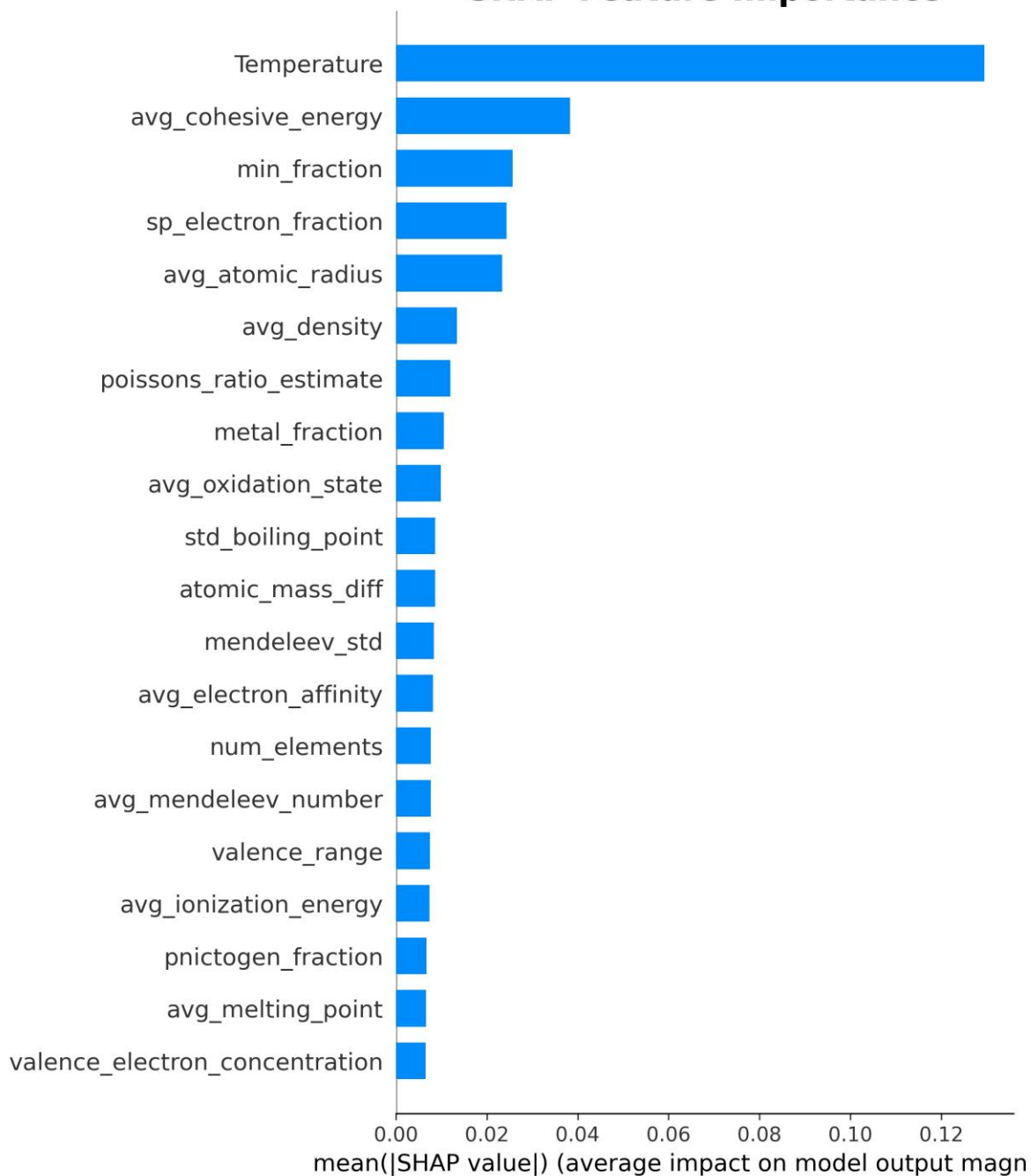
SHAP Summary Plot



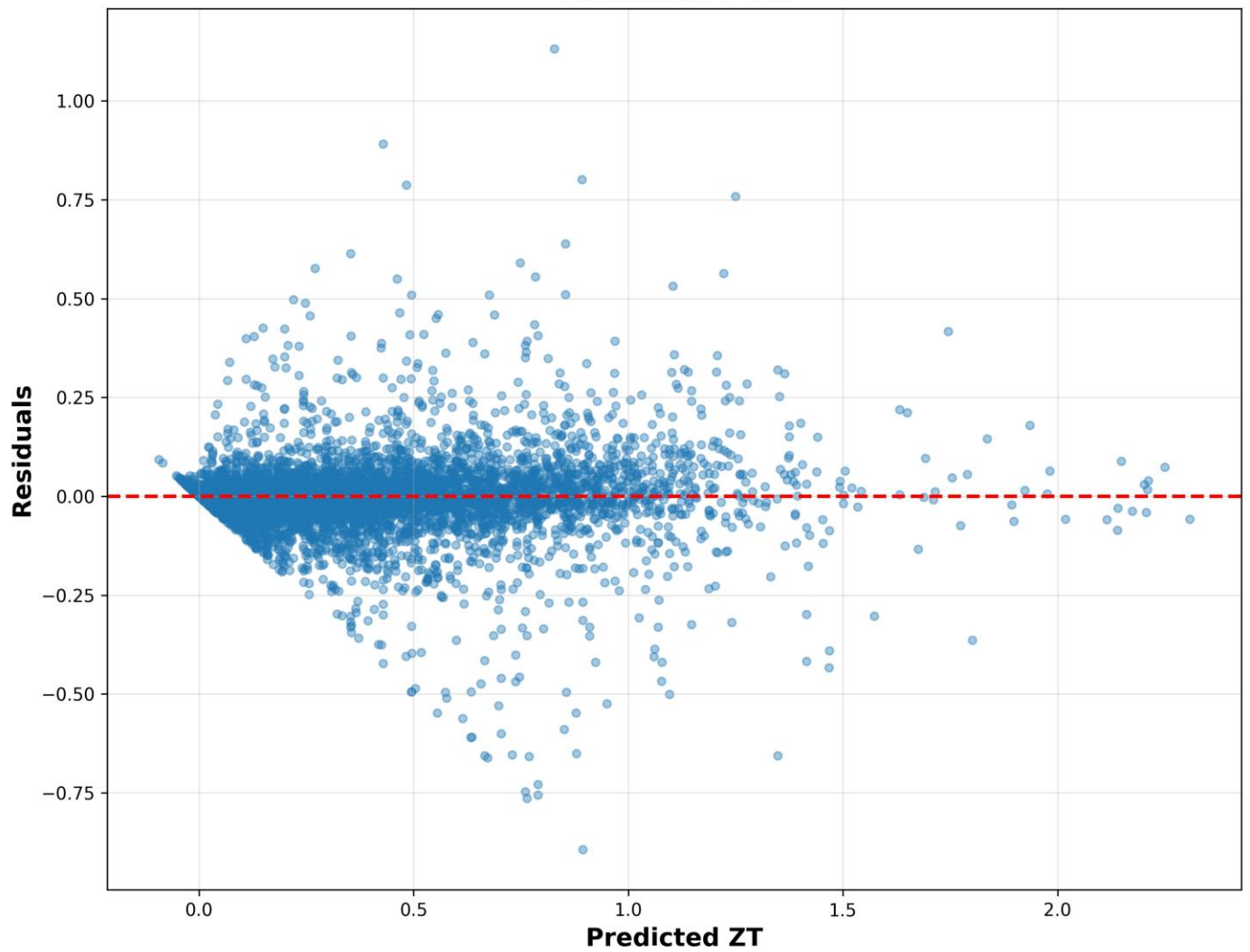
SHAP Dependence Plot : Temperature



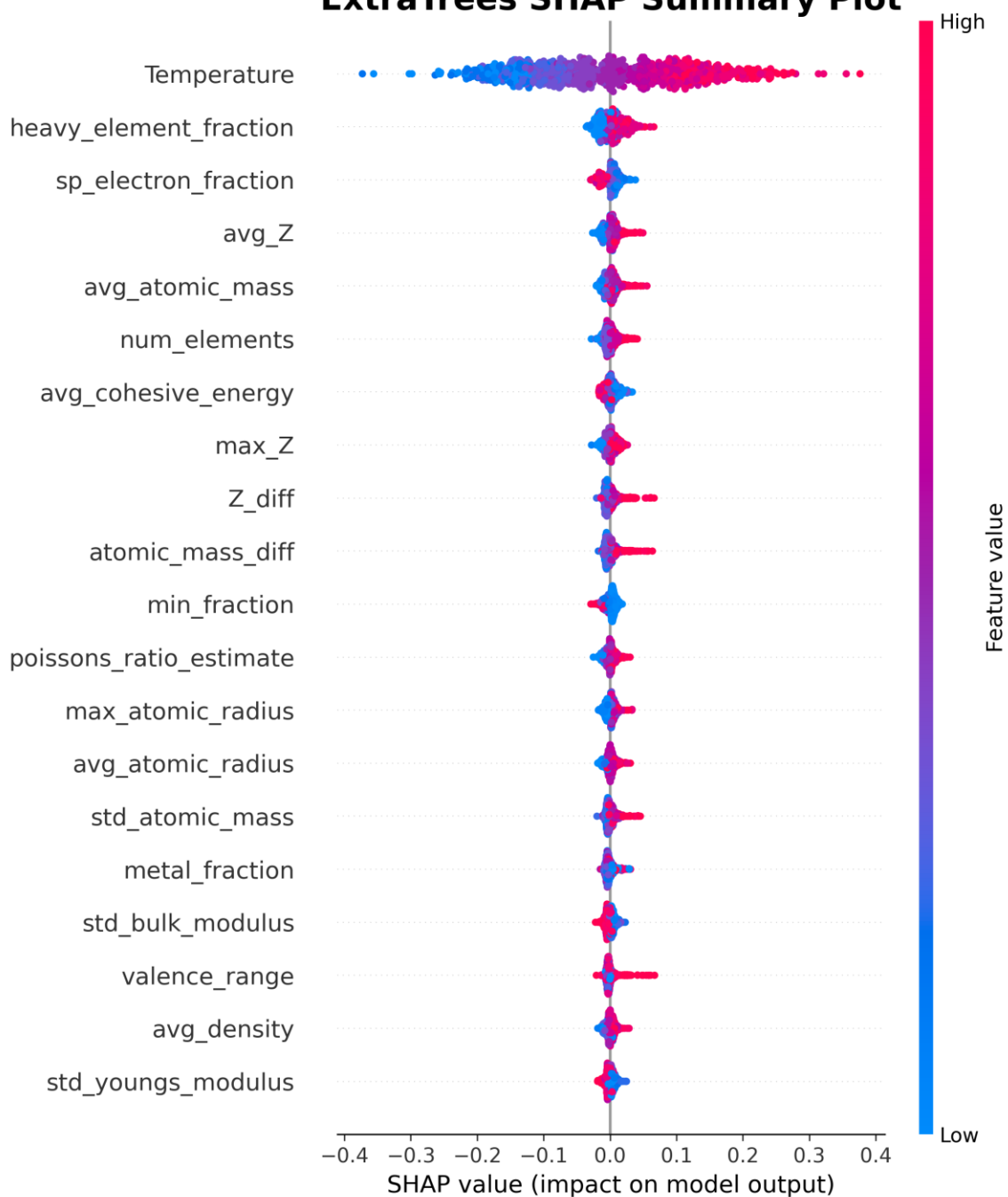
SHAP Feature Importance



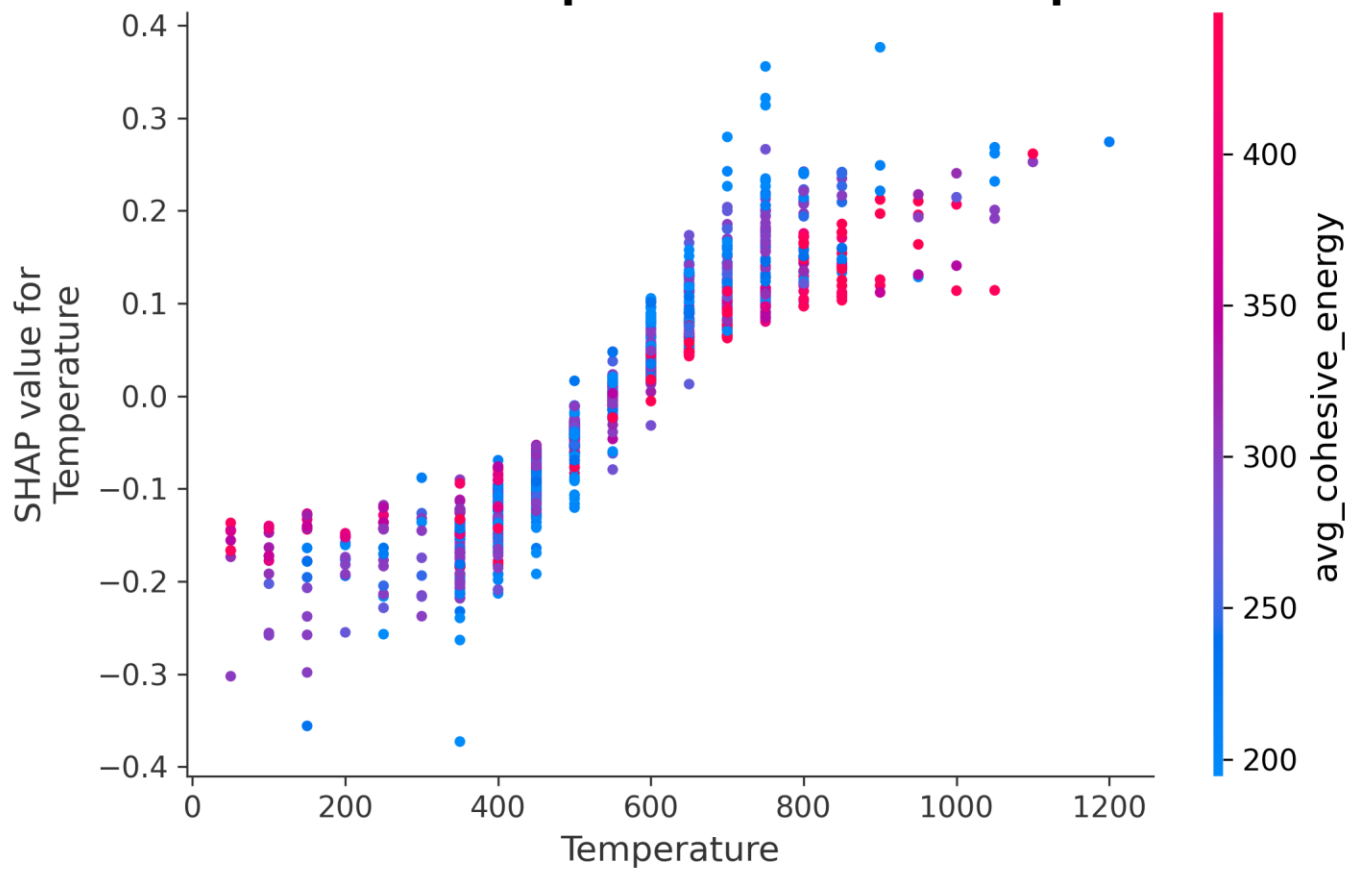
Residual Plot



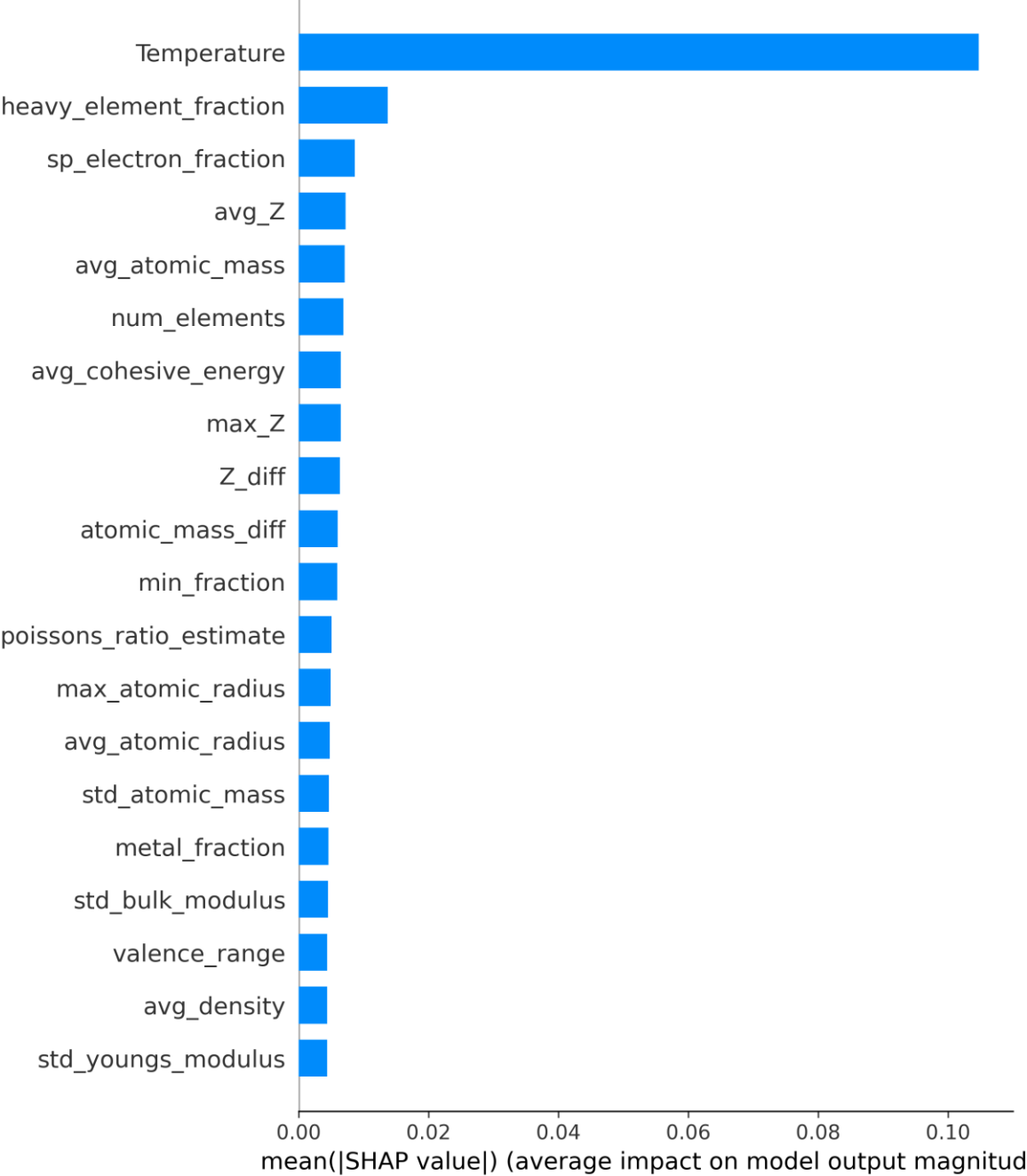
ExtraTrees SHAP Summary Plot



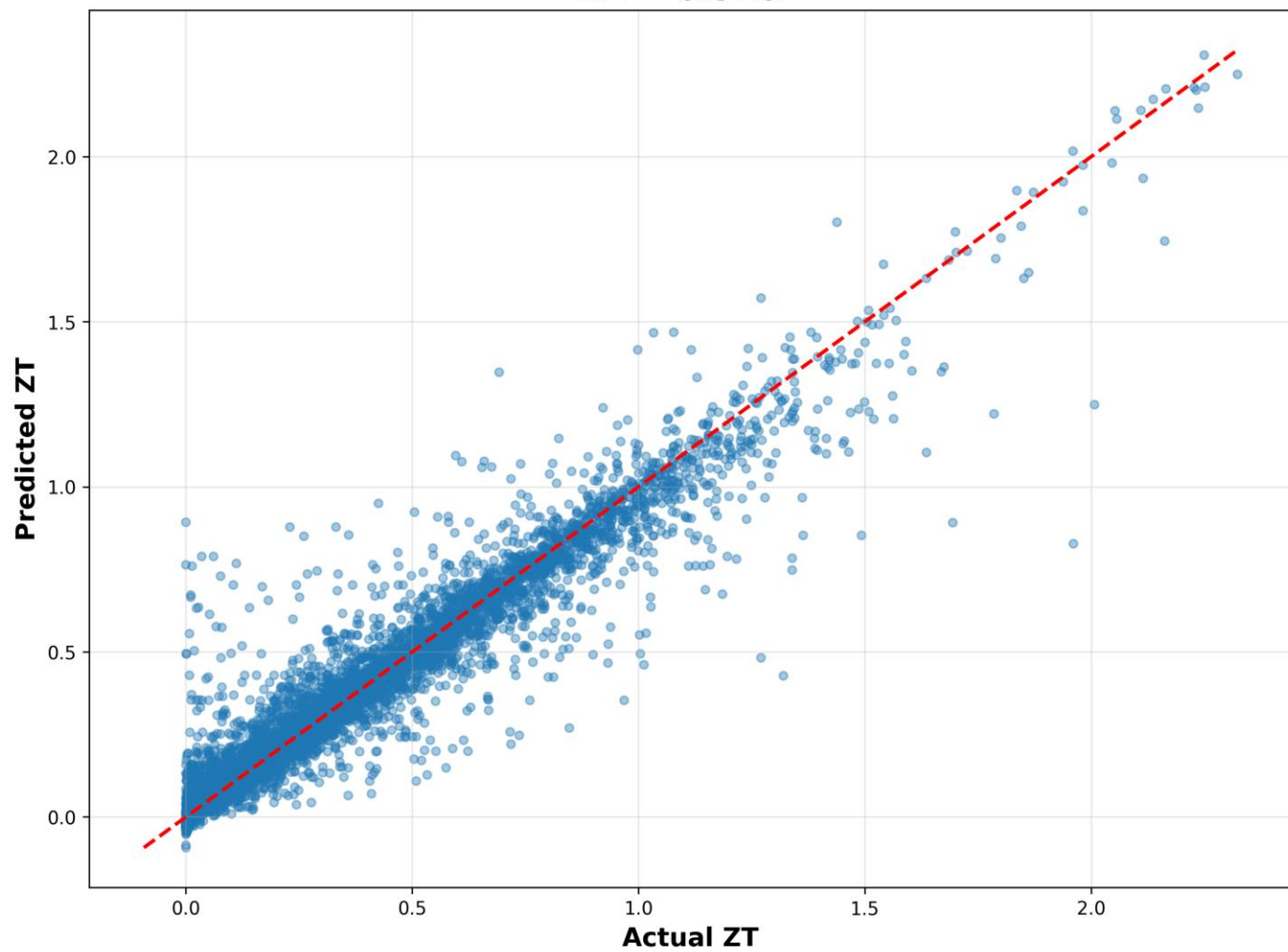
ExtraTrees SHAP Dependence Plot : Temperature

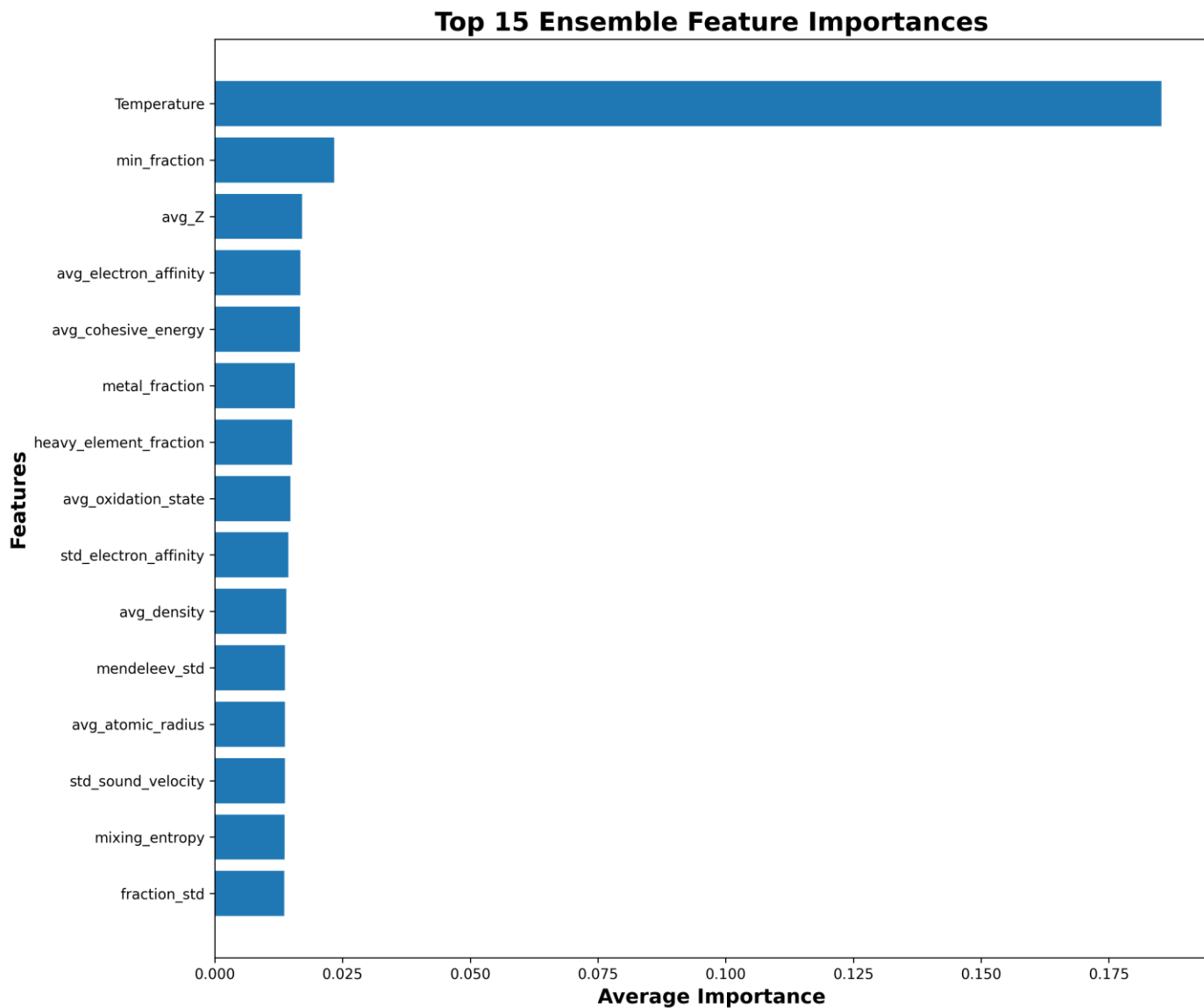


ExtraTrees SHAP Feature Importance



ExtraTrees + LightGBM Ensemble
R2 = 0.9202

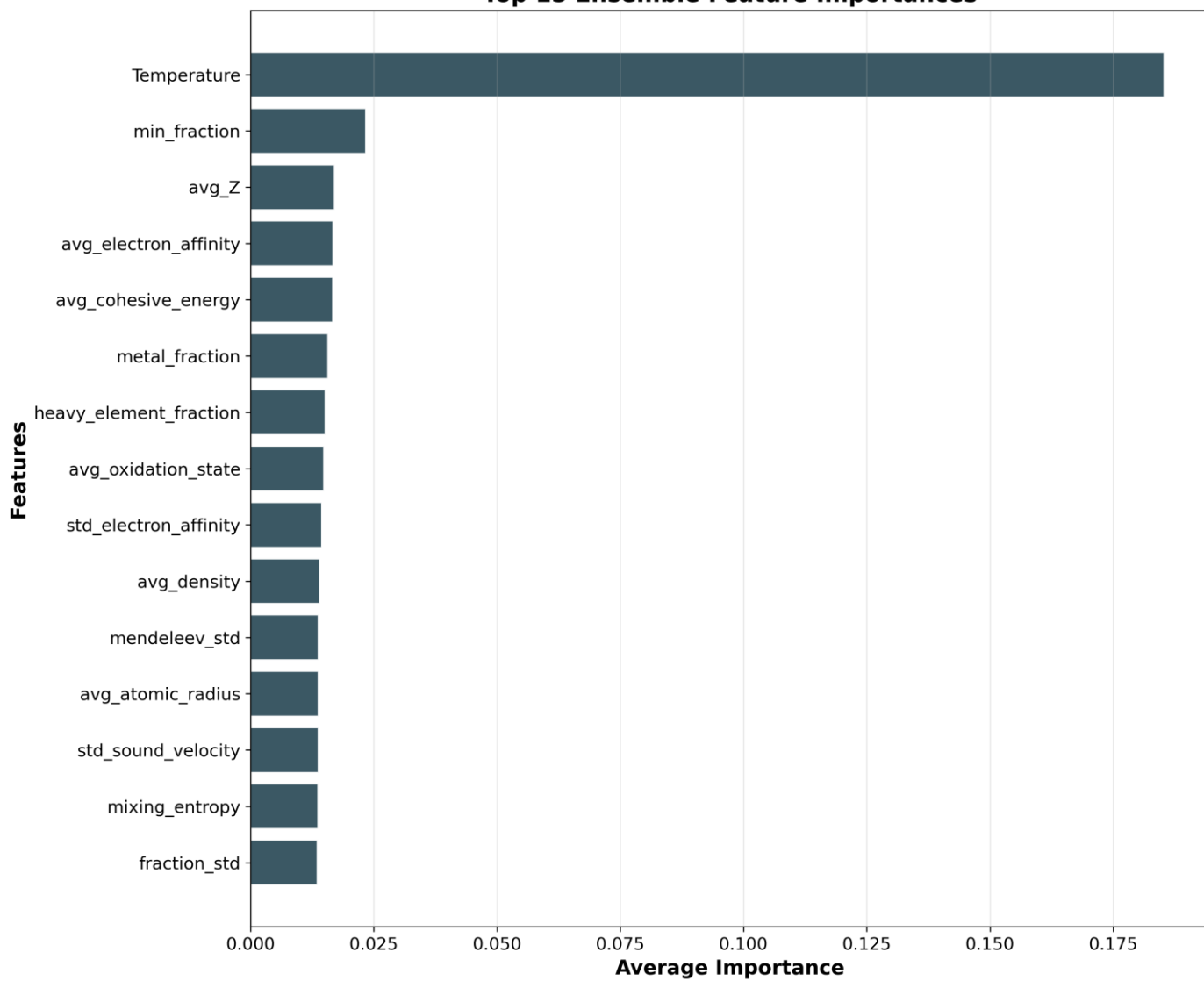




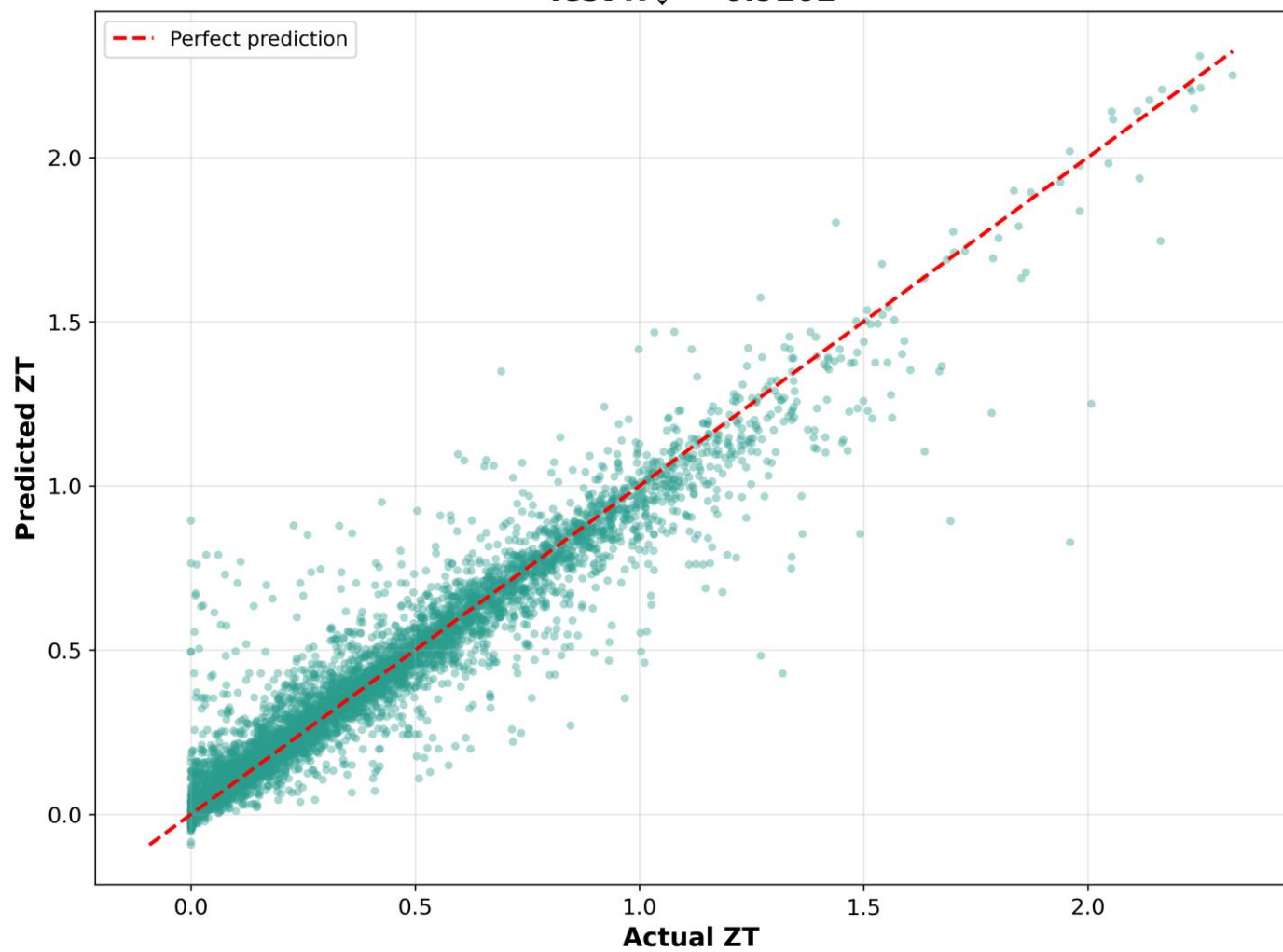
Graphs of Inverse Design:



Top 15 Ensemble Feature Importances

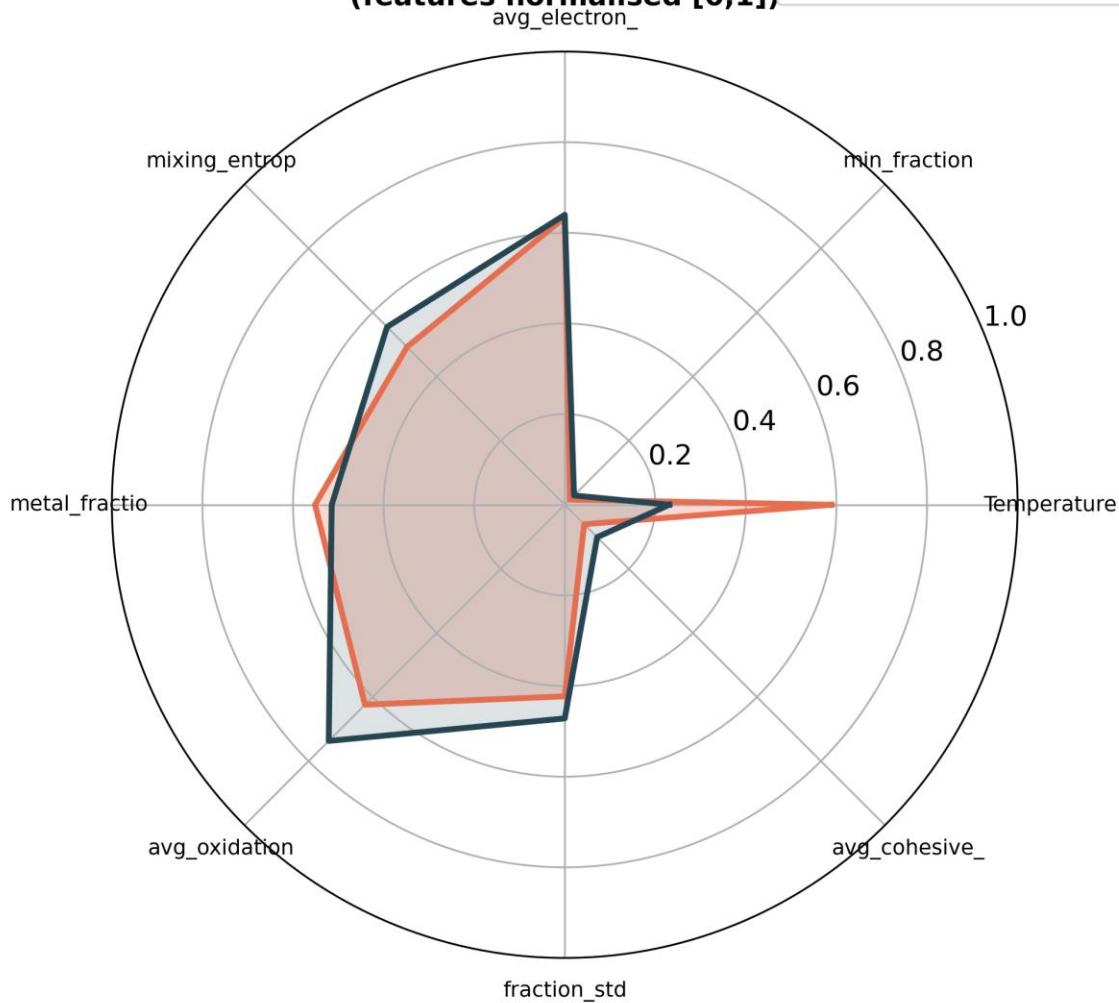


ExtraTrees + LightGBM Ensemble
Test $R^2 = 0.9202$

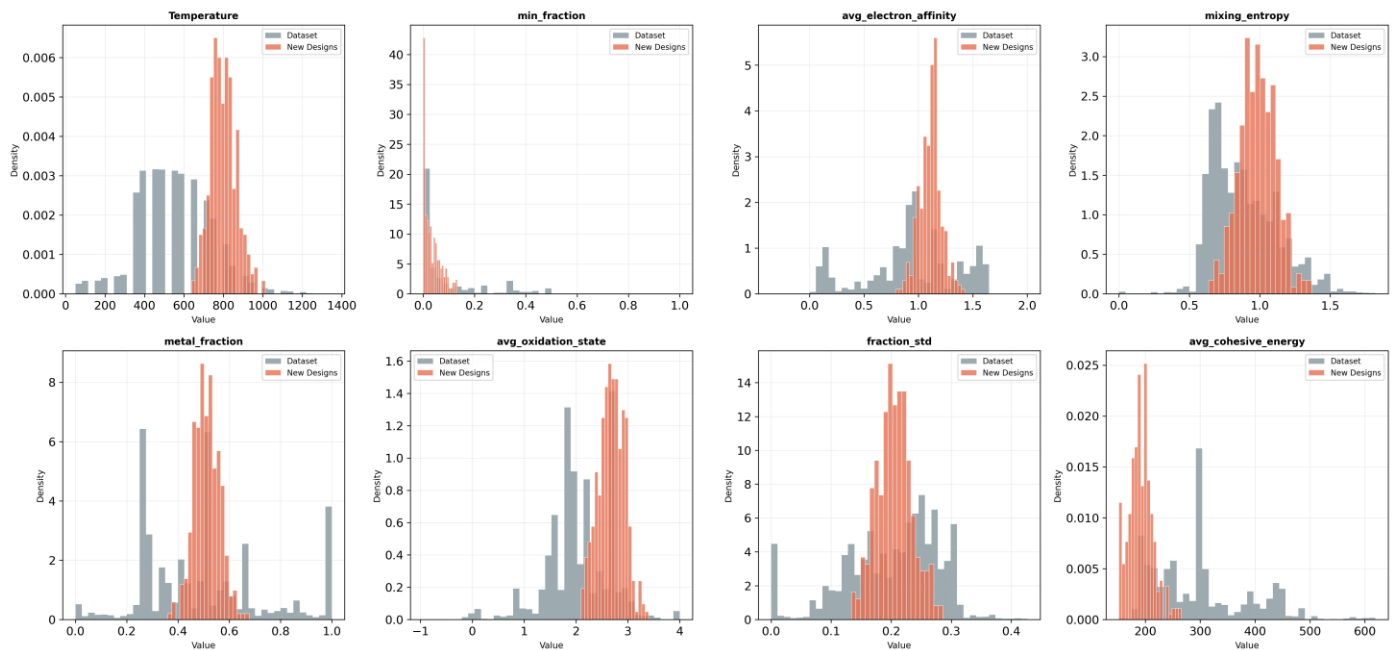


Radar: Best New Design vs Its Seed (features normalised [0,1])

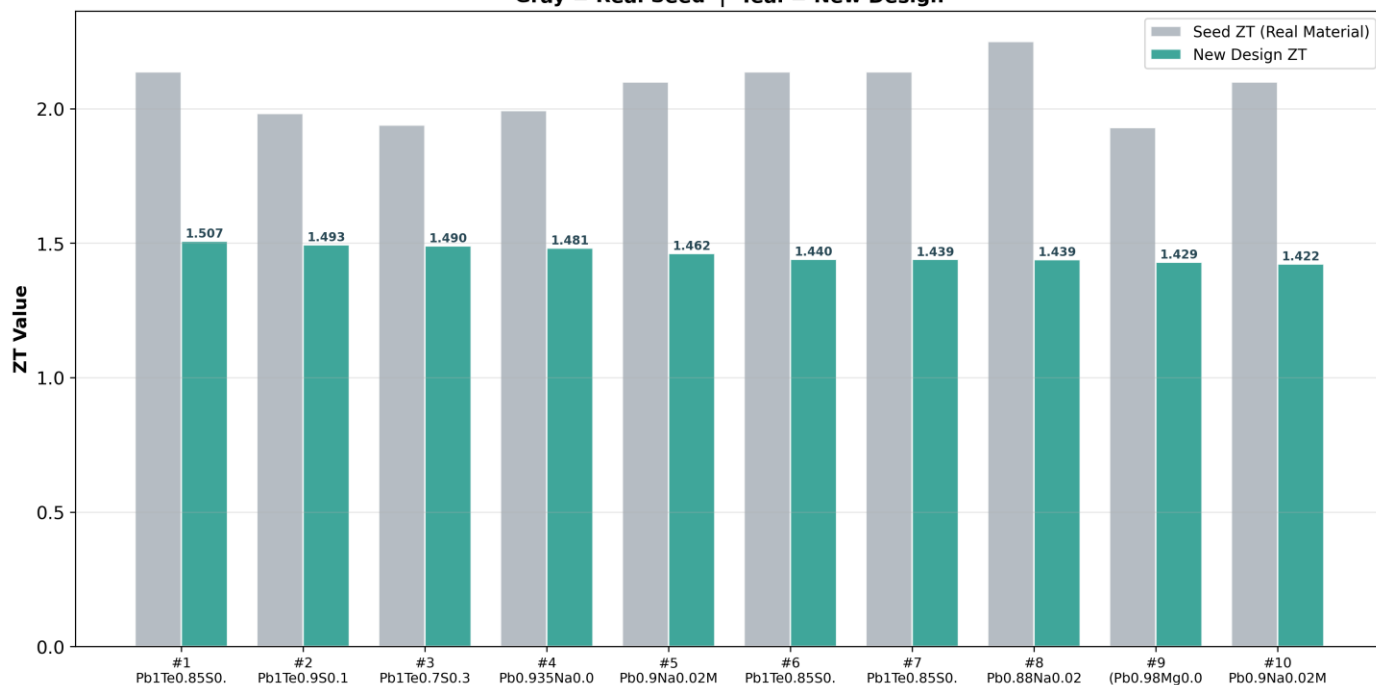
Best New Design ZT = 1.5070
Seed: Pb1Te0.85S0.15Na0.06 ZT = 2.1369



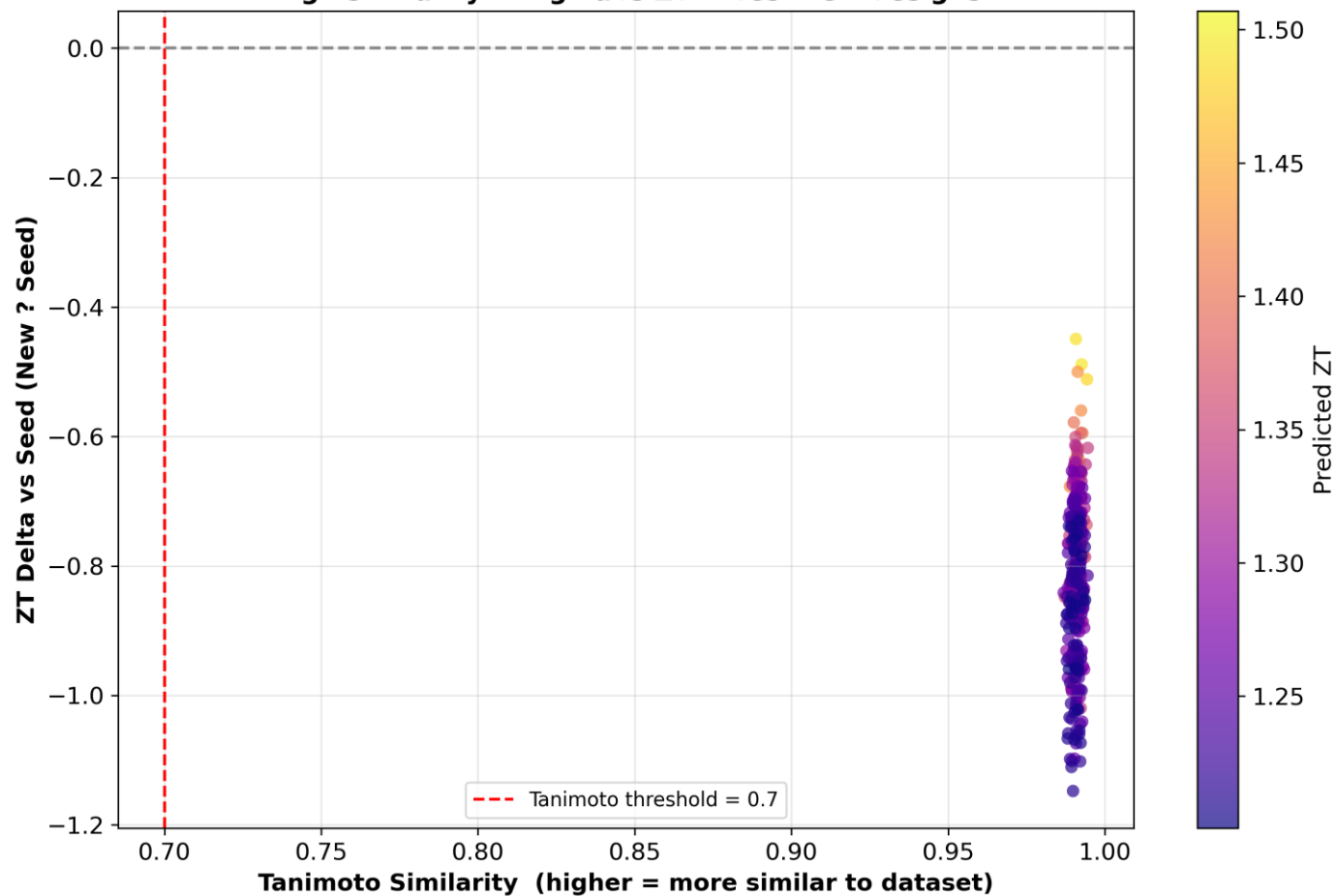
Top 8 Feature Distributions: Dataset vs New Designs
(Overlap = physically grounded designs)

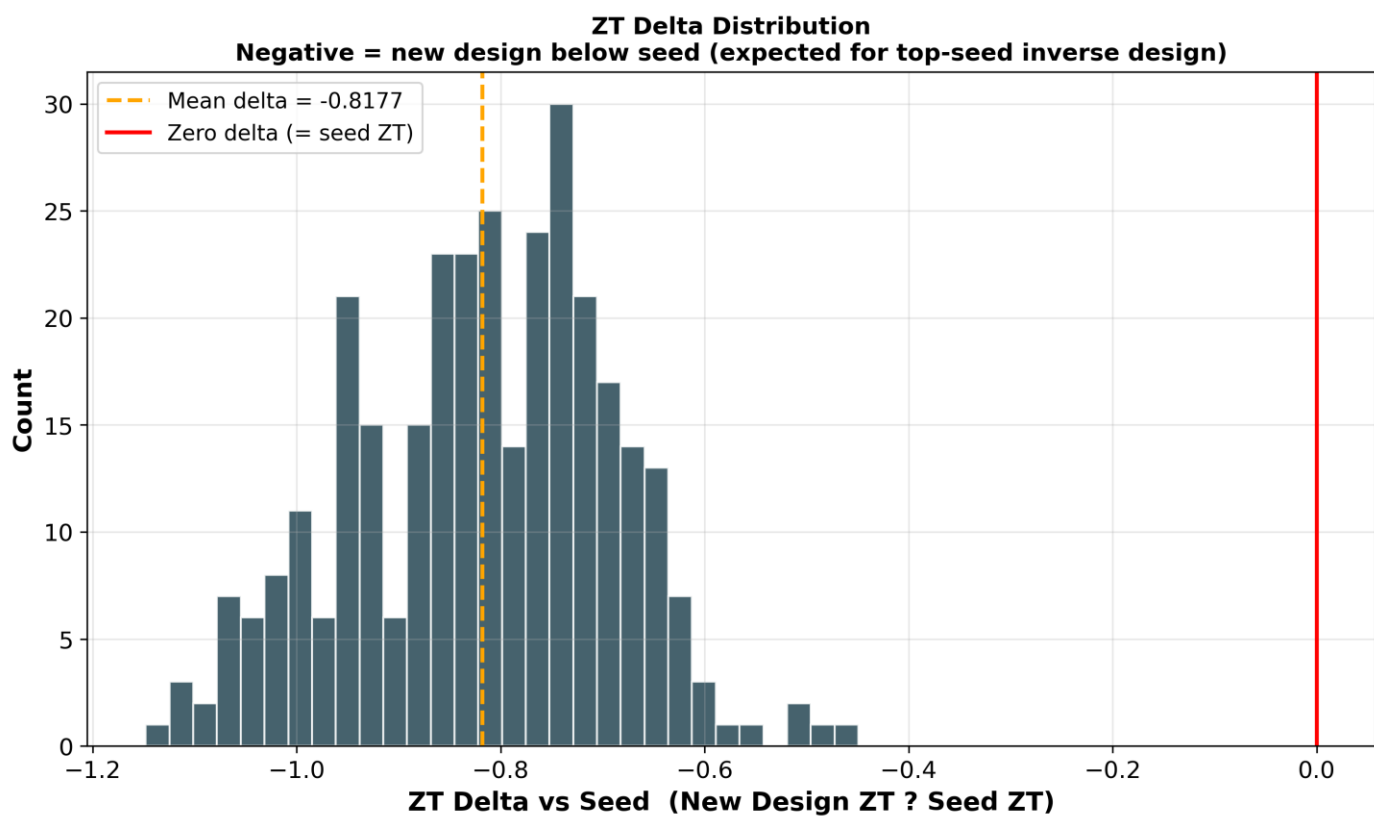
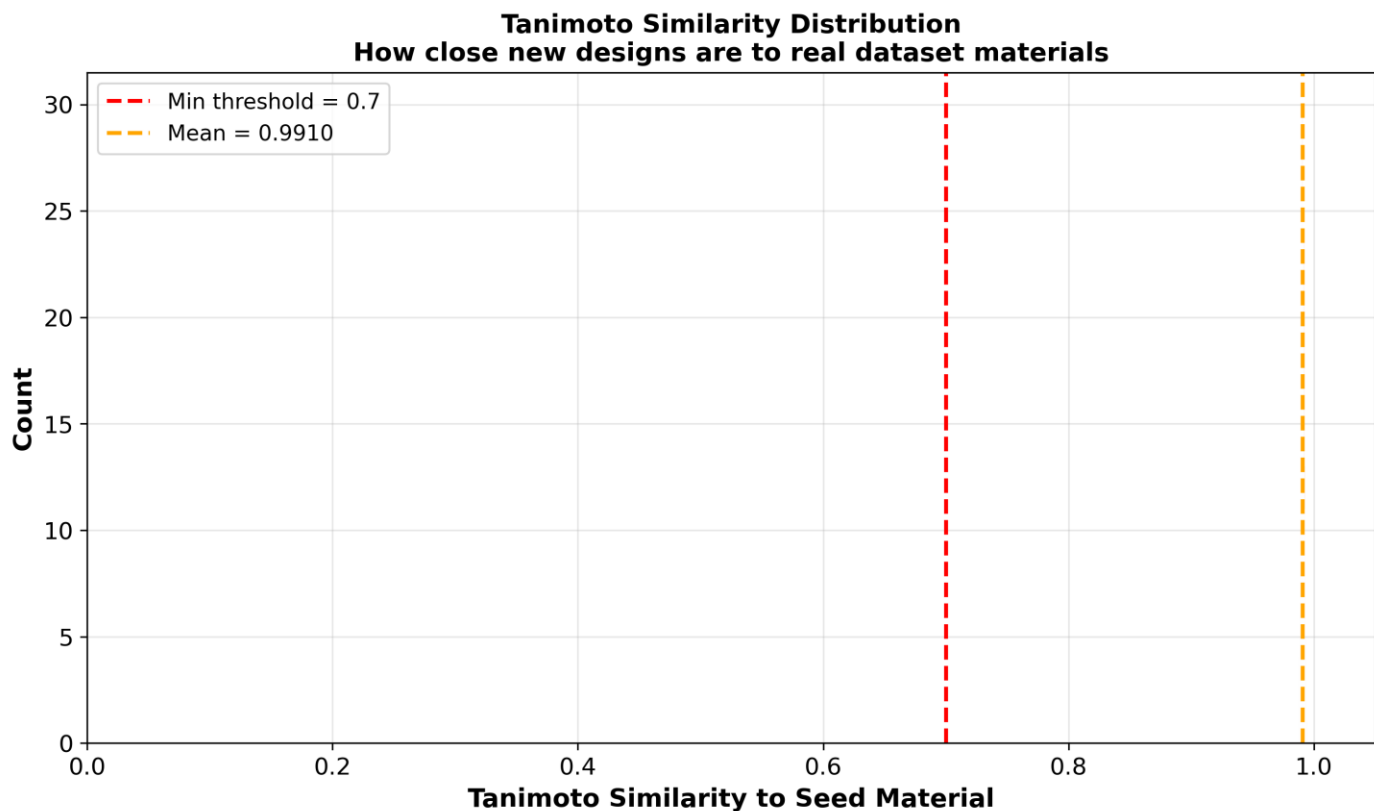


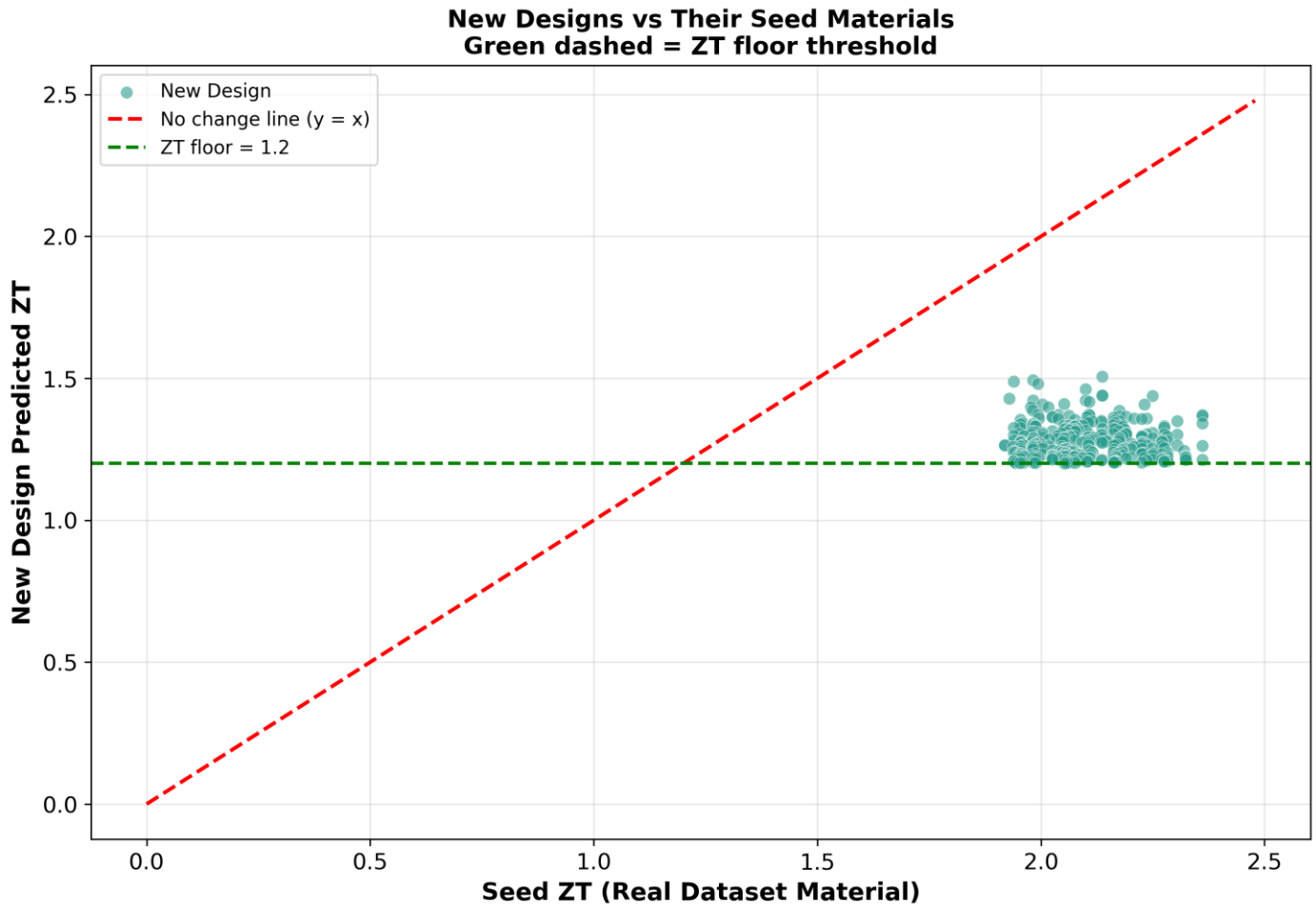
Top 10 New Designs vs Their Seed Materials
 Gray = Real Seed | Teal = New Design



Similarity vs ZT Delta
 High similarity + high abs ZT = best new designs







Discussion:

The comparative analysis reveals detailed information about machine learning models that predict thermo-electric ZT by examining model performance.

This study demonstrates that machine learning and deep learning methods can effectively predict the thermoelectric figure of merit (ZT) using experimental data, including compositional information and physical property measurements.

The research team evaluated several regression algorithms, including CatBoost, Extra Trees, Gradient Boosting Machine (GBM), LightGBM, XGBoost, and ZTNet, a residual deep neural network model, to find the best approach for predicting thermoelectric properties. The ensemble tree-based methods showed better forecasting capabilities as they successfully modeled the complex relationships between thermoelectric material characteristics and their behavior.

The stacked ExtraTrees and LightGBM ensemble achieved its best prediction results through an additional model combination, resulting in the highest test R^2 score of about 0.92. The research indicates that model prediction accuracy improves when multiple models learn complementary skills through their combined strengths. The model prediction errors

followed a normal distribution, The improvement obtained by stacked ensemble learning suggests that thermoelectric ZT prediction benefits from integrating multiple nonlinear representations rather than relying on a single predictor. pattern

around zero, and residual analysis confirmed that these errors did not show any specific pattern across all samples.

Feature-importance and SHAP analyses consistently highlighted temperature as the main factor influencing thermoelectric performance results.

This finding is physically significant, as thermoelectric transport properties depend on temperature-related carrier transport and phonon scattering methods. The study also identified other key factors that affect electron affinity and cohesive energy. The inverse-design workflow shows.

Novelty and Contributions:

The primary contribution of this work is not the proposal of a new machine learning architecture but the integration of benchmarking, stacked ensemble prediction, model interpretation, and constrained inverse design into a unified thermoelectric screening workflow.

The proposed framework:

1. Benchmarks multiple ML and DL approaches for ZT prediction.

2. Introduces stacked ensemble evaluation.
3. Uses SHAP for physical interpretation.
4. Extends prediction toward similarity-guided inverse candidate generation.

Main contributions:

1. Developed a composition-driven ML workflow for ZT prediction.
2. Benchmarked multiple ML and DL approaches.
3. Proposed stacked ensemble learning for improved accuracy.
4. Introduced similarity-guided inverse design.
5. Applied SHAP for model interpretability.

Limitations, Future Work and Conclusion:

There were some limitations such as the size of dataset it is way more large for ML models, and contains so much of experimental noise, dataset doesn't contain any crystal-descriptors, couldn't apply SHAP to integrative model because they are

Stacked ensemble models , these predictions still require experimental validation

Future work will focus on integrating graph neural networks, DFT-derived descriptors, active learning pipelines and experimental validation of generated candidate materials.

The present study demonstrates that machine learning and inverse design methodologies can effectively predict and explore thermoelectric materials with high accuracy and strong physical consistency.

References :

- Sharma, V., Singh, R., & co-authors. (2022). Large dataset-driven machine learning models for accurate prediction of thermoelectric figure of merit (zT). *ACS Applied Materials & Interfaces*, 14, 4
- Wang, Y., Zhang, H., & co-authors. (2023). Machine learning for predicting ZT values of high-performance thermoelectric materials. *APL Materials*, 11(8),
- Li, X., Chen, Y., & co-authors. (2024). Machine-learning-guided prediction of thermoelectric properties from chemical formula. *Journal of Materials Chemistry C*, 12, 2024
- Kumar, A., Gupta, S., & co-authors. (2024). Physics-informed neural networks for prediction of thermoelectric transport properties. *Computational Materials Science*, 240, 113012.
- Zhao, J., Liu, P., & co-authors. (2024). Graph neural network enabled discovery of high-performance thermoelectric materials. *npj Computational Materials*, 10, Article 145.
- Patel, D., Roy, K., & co-authors. (2025). Thermoelectric material performance (zT) predictions with machine learning using large-scale experimental datasets. *ACS Applied Materials & Interfaces*, 17, 2025.
- Sato, T., Tanaka, H., & co-authors. (2025). Active learning assisted discovery and optimization of thermoelectric materials. *Advanced Functional Materials*, 35, 2501245.
- Nguyen, M., Lee, J., & co-authors. (2025). Inverse design framework for high-ZT thermoelectric materials using deep generative models. *Nature Computational Science*, 5, 312-324.
- Huang, Z., Wang, C., & co-authors. (2025). Explainable artificial intelligence for thermoelectric materials design and ZT prediction. *Scientific Reports*, 15, Article 44723.
- Garcia, E., Smith, R., & co-authors. (2025). Generative artificial intelligence for thermoelectric material discovery and optimization. *Materials Horizons*, 12, 1450-1464.
- Wide-Temperature-Range n-Type $\text{Mg}_3(\text{Sb,Bi})_2$ (Nature Communications, 2023)
- GeTe-Based Thermoelectrics with Defect Engineering (Nature Communications, 2022)
- Data-Driven Enhancement of ZT in SnSe-Based Systems (JACS, 2022)

